Progress at the 5th post-Minkowskian Order in Gravitational-Wave Physics

Zvi Bern

June 18, 2026

Theoretical Developments of Gravitational Wave Physics

Yukawa Institute, Kyoto

ZB, E. Herrmann, R. Roiban, M. Ruf, V.A. Smirnov, A.V. Smirnov, M. Zeng: arXiv:2305.08981, arXiv:2406.01554; arXiv:2509.17412

ZB, E. Herrmann, R. Roiban, M. Ruf, A.V. Smirnov, S. Smith, M. Zeng: arXiv:2512.15383.

ZB, Jackman, Mansfield, Ruf: ArXiv:2603.15283.

UCLA Mani L. Bhaumik Institute for Theoretical Physics

Funded by the European Union

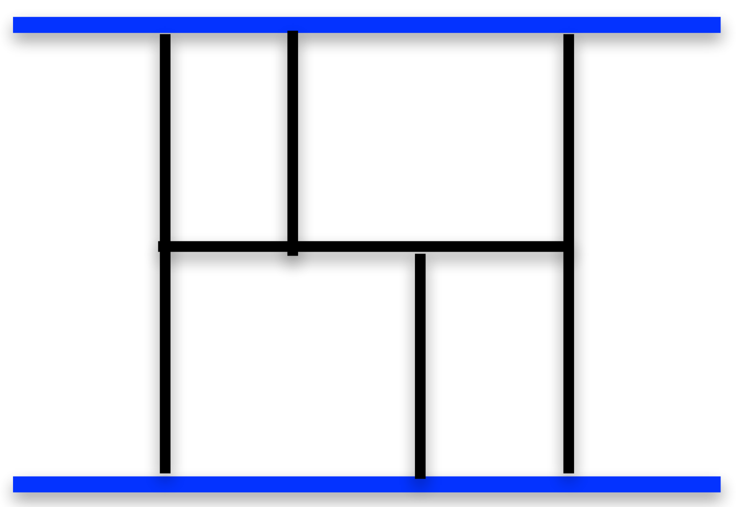
European Research Council
Established by the European Commission

Outline

1. Summary of the problem and PM approach.

2. Brief review of basics:
   - Quantum scattering amplitudes and gravity.
   - Unitarity as a tool.
   - Double copy relations between gravity and gauge theory.

2. Applications to state-of-the-art precision gravitational-wave computations.

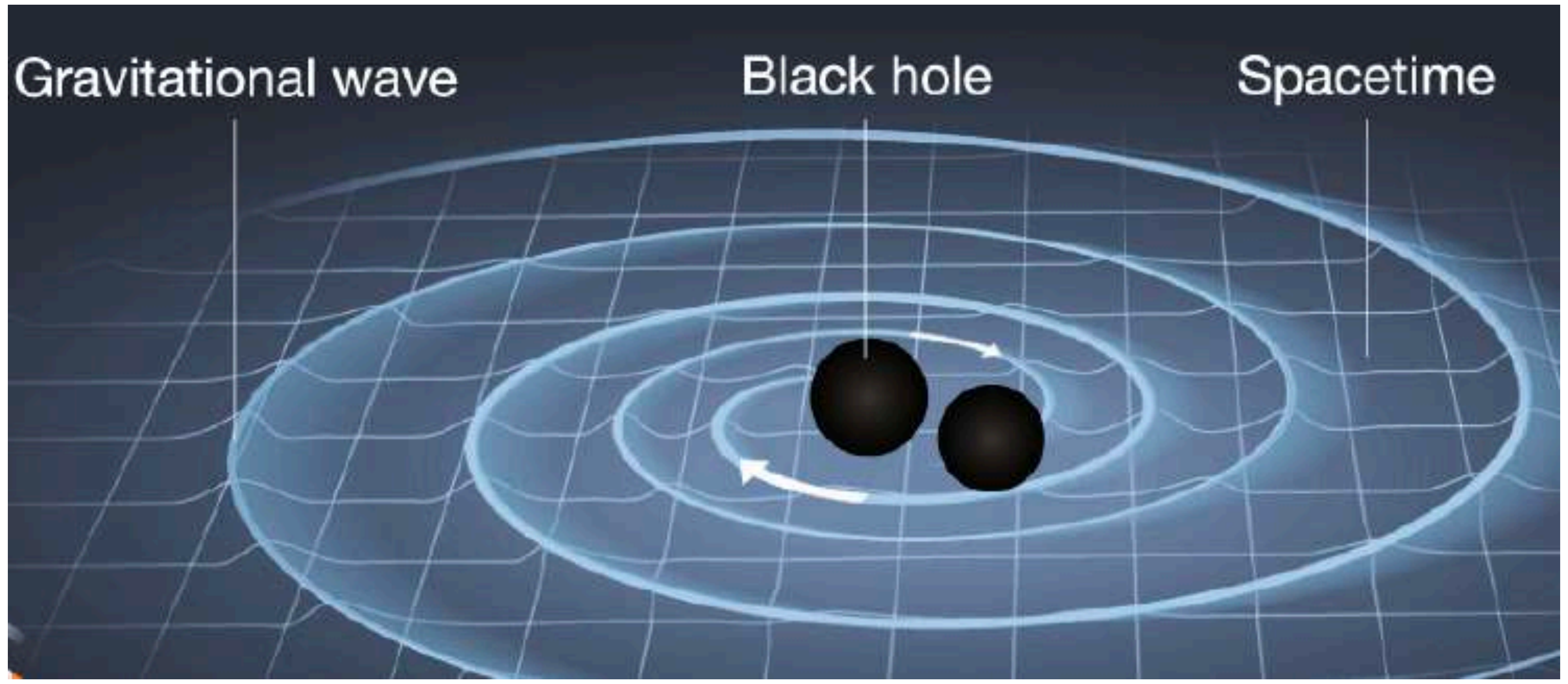


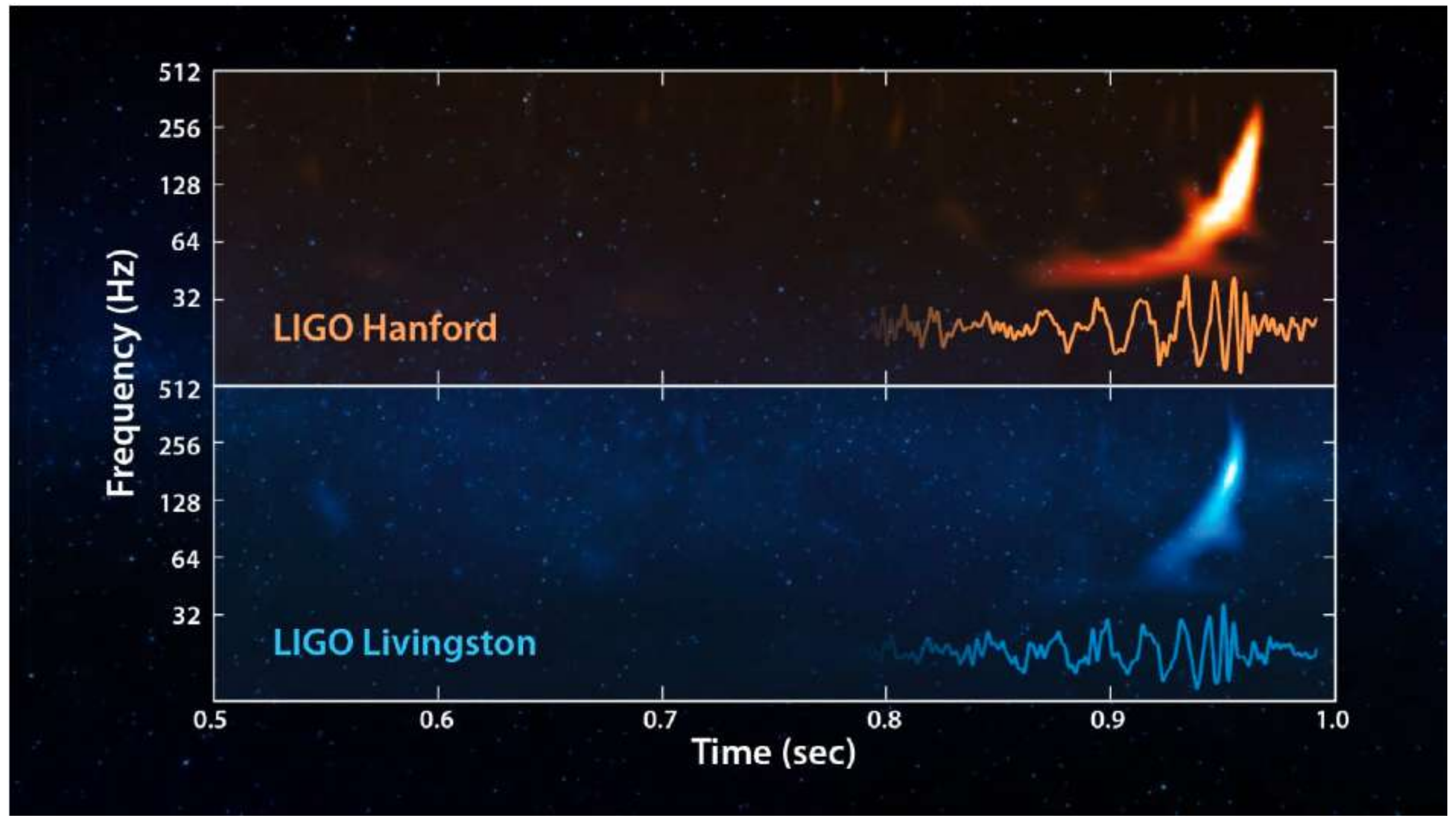
3. Update on $O(G^5)$

4. Conclusions and outlook.

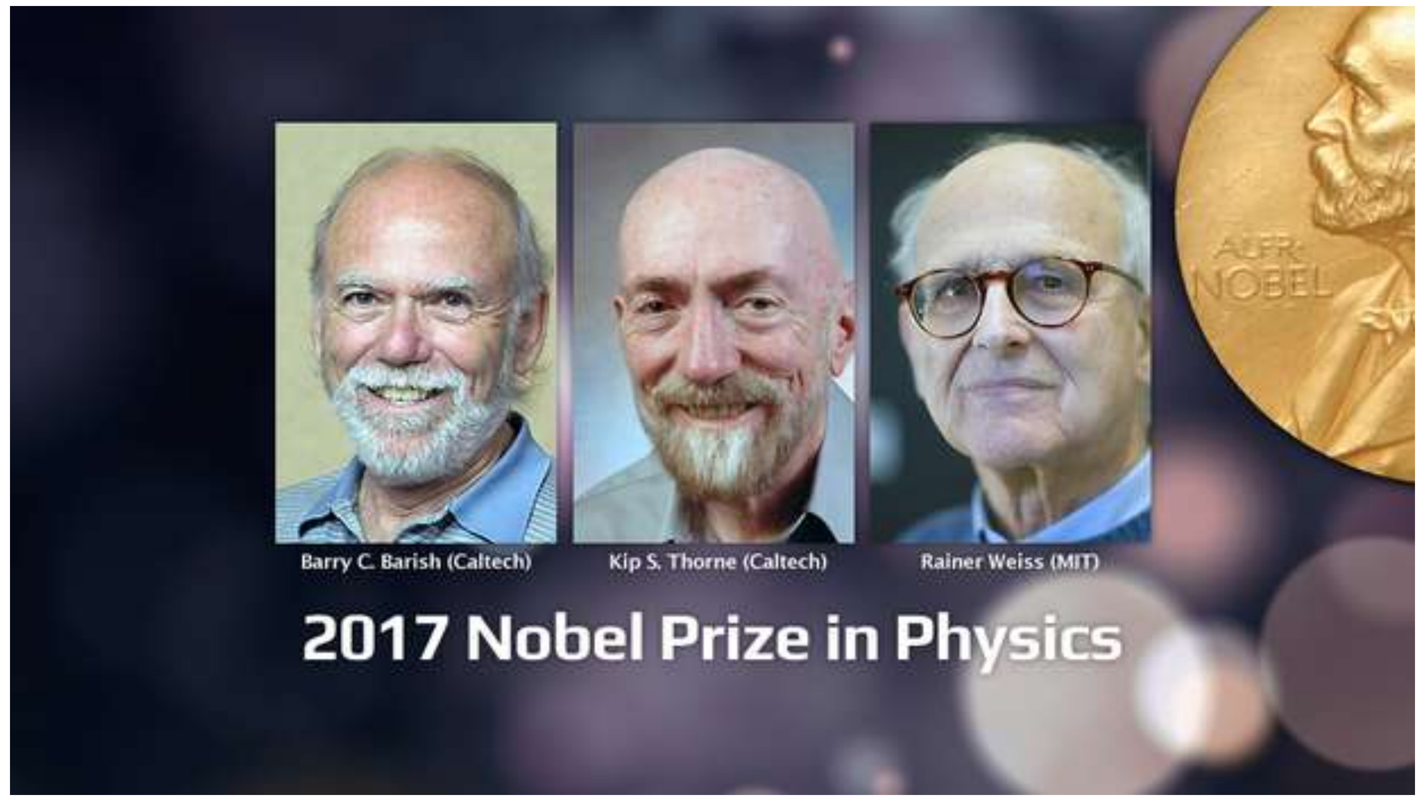
Gravitational Waves

Era of gravitational-wave astronomy



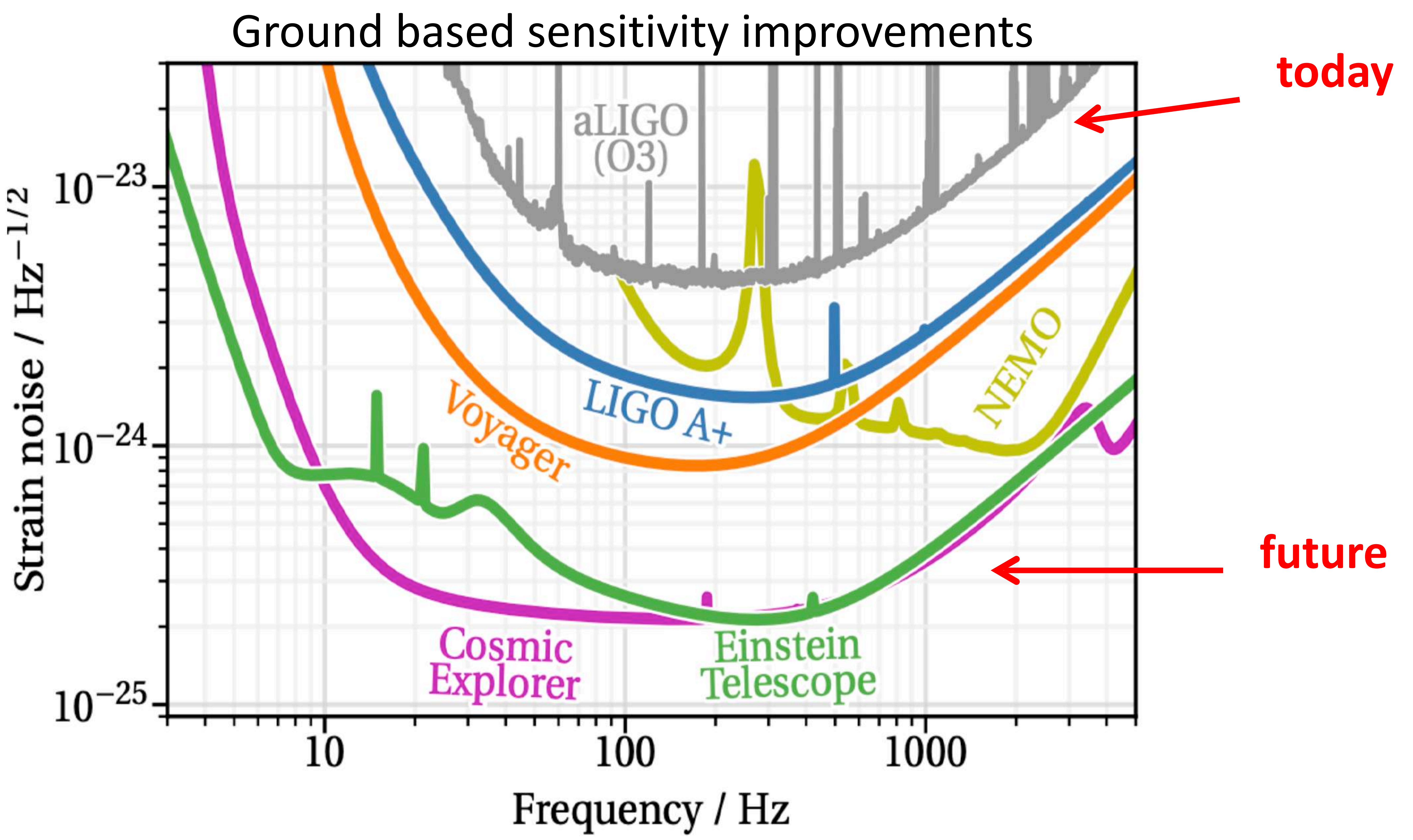


LIGO Detectors



How can we, as particle physicists, help?

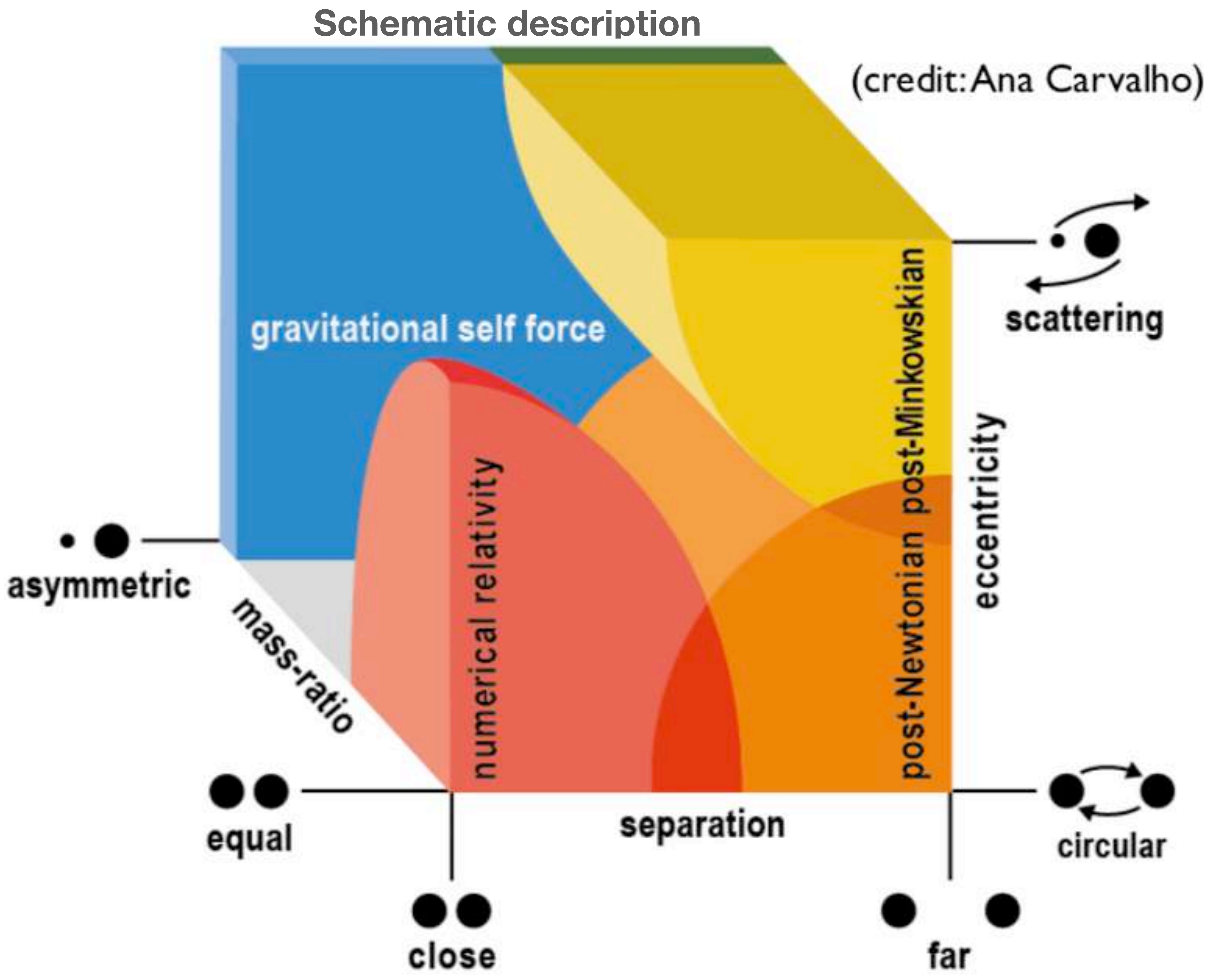
Increased Sensitivity of Detectors



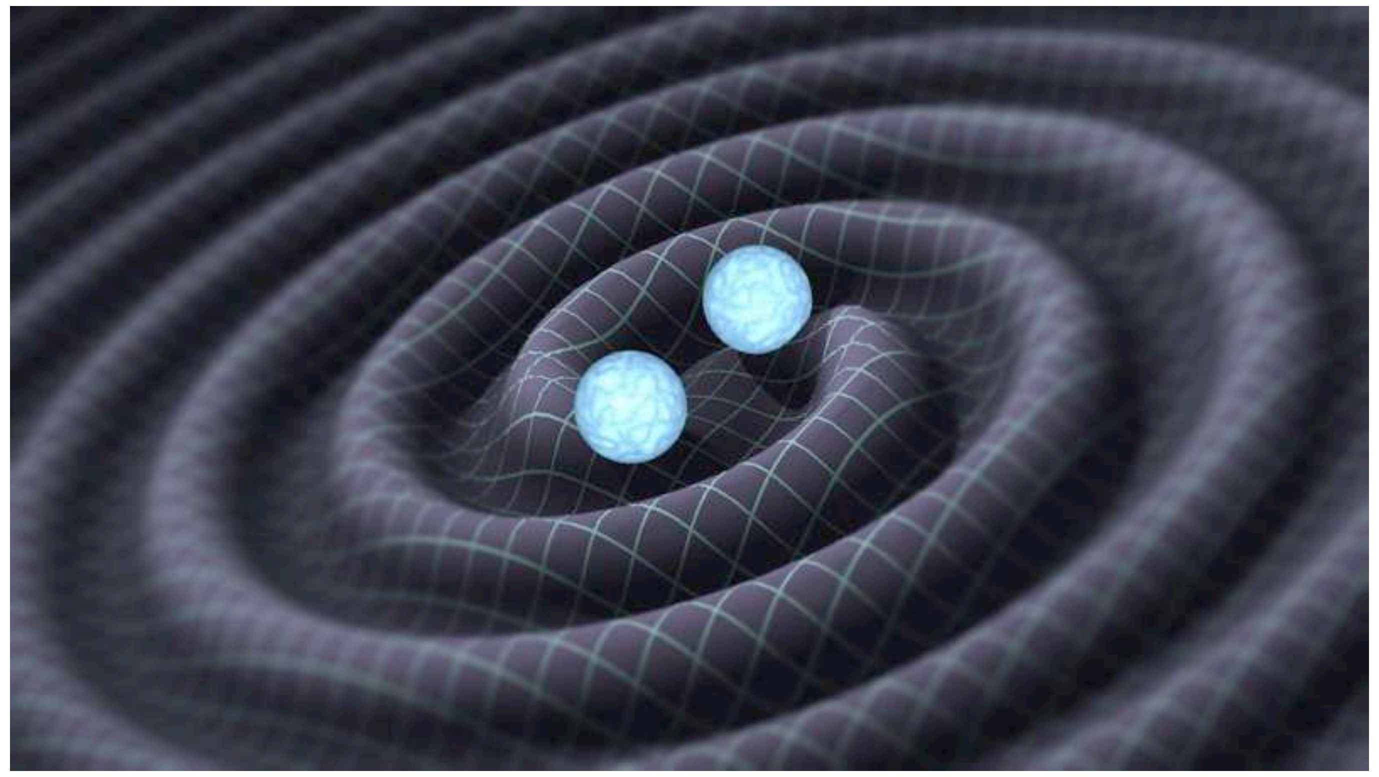
https://cosmicexplorer.org/sensitivity.html

- **Factor of 100 increase in sensitivity in some regions is coming.**
- **Also, space-based detectors at much lower frequencies.**
- **Highly nontrivial theory challenge. Two more orders in perturbation theory.**

Post-Minkowskian (PM) Approach



Scattering amplitudes fits naturally with post-Minkowskian approach

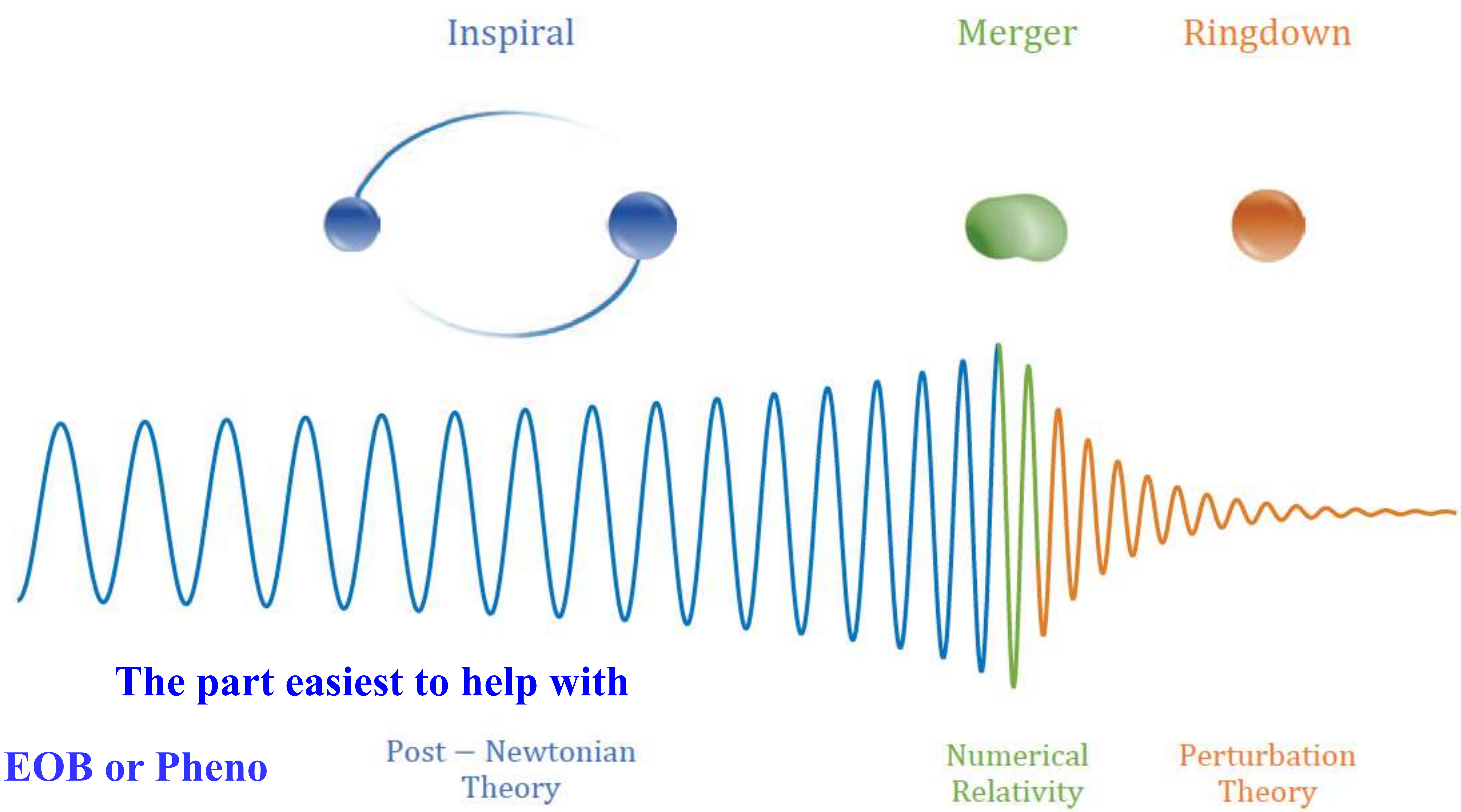


Many talks about the different approaches.

Different approximations are valid in different regions.
PM is an expansion in G

Information needs to be combined into a unified (EOB) waveform model.

Goal is to Improve Waveforms



The part easiest to help with

PN + EOB or Pheno

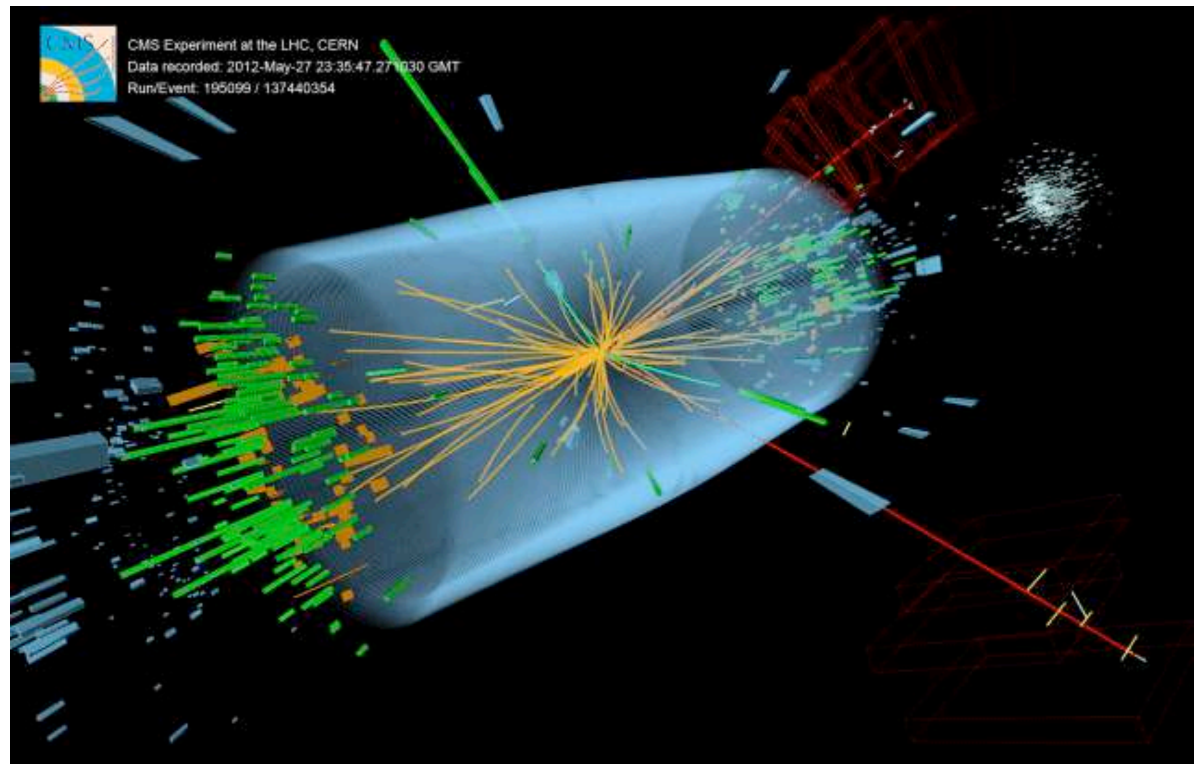
From Antelis and Moreno, arXiv:1610.03567

Small errors accumulate. Need for high precision.

What do QFT Amplitudes have to do with Gravitational Waves?

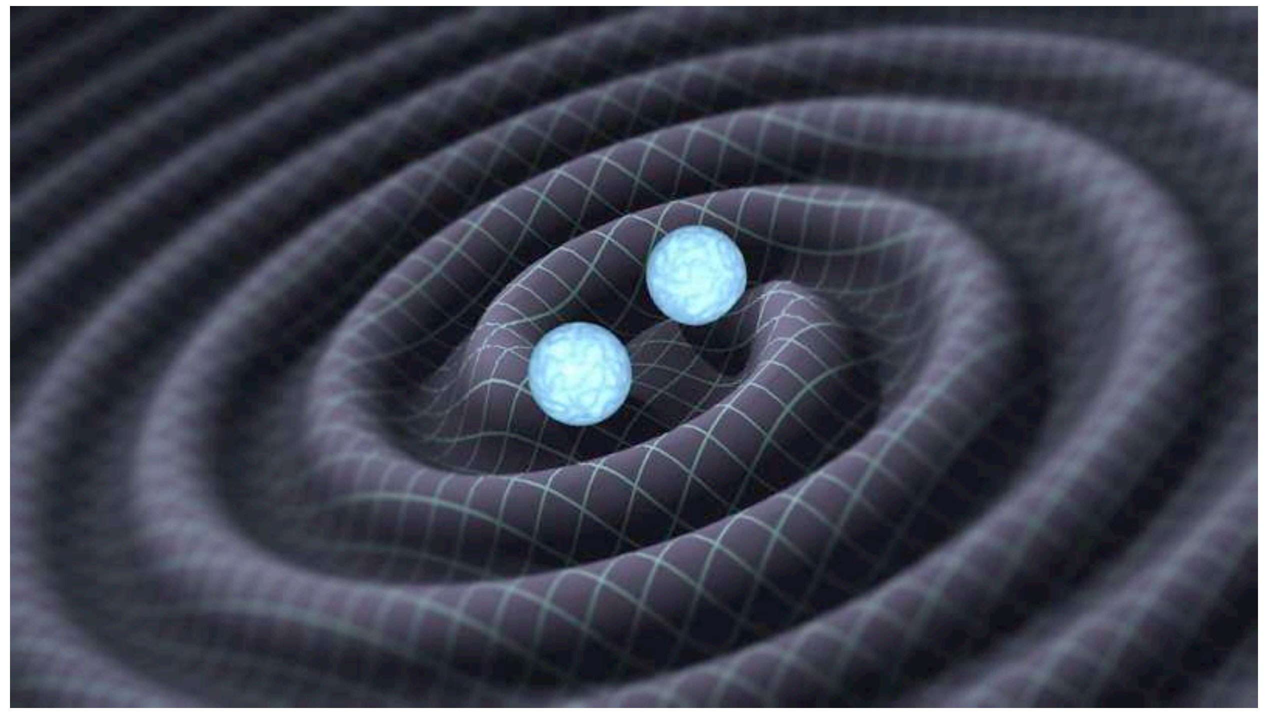
What does particle physics have to do with classical dynamics of astrophysical objects?

unbounded trajectory



gauge theories, QCD, electroweak
quantum field theory

bound orbit



General Relativity
classical physics

Effective field theory ideas: Black holes and neutron stars are point particles as far as long-wavelength radiation is concerned.

Iwasaki (1971); **Goldberger, Rothstein (2006)**; Neill and Rothstein; Vaydia, Foffa , Porto, Rothstein, Sturani; Kol; Bjerrum-Bohr, Donoghue, Holstein, Plante, Pierre Vanhove; Levi, Steinhoff; Vines etc

Post-Minkowskian Approach

PM has a long history. Some highlights:

- **Kerr,** *The Lorentz-covariant approximation method in general relativity* **(1959)**
- **Bertotti and Plebanski,** *Theory of gravitational perturbations in the fast motion approximation* **(1960)**
- **Kovacs and Thorne,** *The generation of gravitational waves* **(1977)**
- **Westpfahl and Goller,** *Gravitational scattering of two relativistic particles in post-linear approximation* **(1979)**
- **Bel, Damour, Deruelle, Ibañez, Marti,** *Poincaré-invariant gravitational field and equations of motion of two pointlike objects: the postlinear approximation of general relativity* **(1981)**
- **Blanchet, Damour,** *Méthode d'itération post-minkowskienne et structure des champs gravitationnels radiatifs* **(1984)**
- **Damour,** *Gravitational scattering, post-Minkowskian approximation and effective one-body theory* **(2016)**
- **Damour,** *High energy gravitational scattering and the general relativistic two-body problem* **(2017)**

In 2018 Modern scattering amplitudes methods brought to bear on the problem. First 3PM paper:

- **Bern, Cheung, Roiban, Shen, Solon, Zeng,** *Scattering amplitudes and the conservative Hamiltonian for binary systems at third Post-Minkowskian order* **(2019)**

Since then many PM papers: 3PM, 4PM, 5PM in progress.
Topic of this talk.

High-energy gravitational scattering and the general relativistic two-body problem

Thibault Damour*
Institut des Hautes Etudes Scientifiques, 35 route de Chartres, 91440 Bures-sur-Yvette, France
(Dated: October 31, 2017)

"… and we urge amplitude experts to use their novel techniques to compute the 2-loop scattering amplitude of scalar masses, from which one could deduce the third post-Minkowskian effective one-body Hamiltonian."

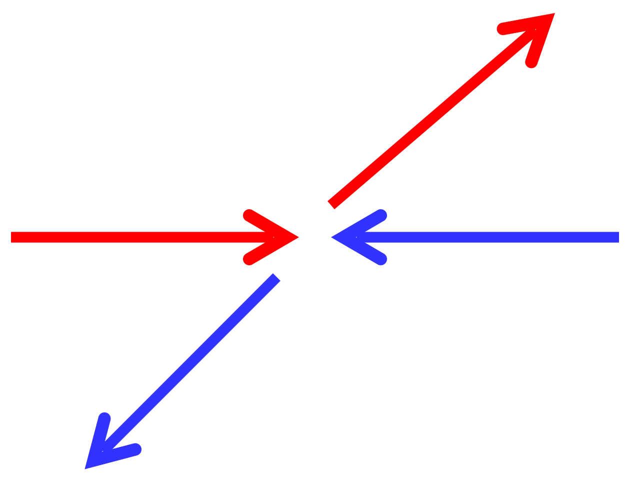
tum gravitational sc… two particles, and we urge amplitude experts to use their novel techniques to compute the 2-loop scattering amplitude of scalar masses, from which one could deduce the third post-Minkowskian effective one-body Hamiltonian

- **August 2017, Alessandra Buonanno spoke at Current Themes in High Energy Physics and Cosmology Conference at Bohr Institute.**
- **December 2017, Alessandra came to QCD Meets Gravity at UCLA, to teach us about gravitational waves.**
- **June 2018, Alessandra came to Amplitudes 2018 at SLAC, again emphasizing 3 PM.**

PM is a natural arena for particle physics: "Pertubation theory"

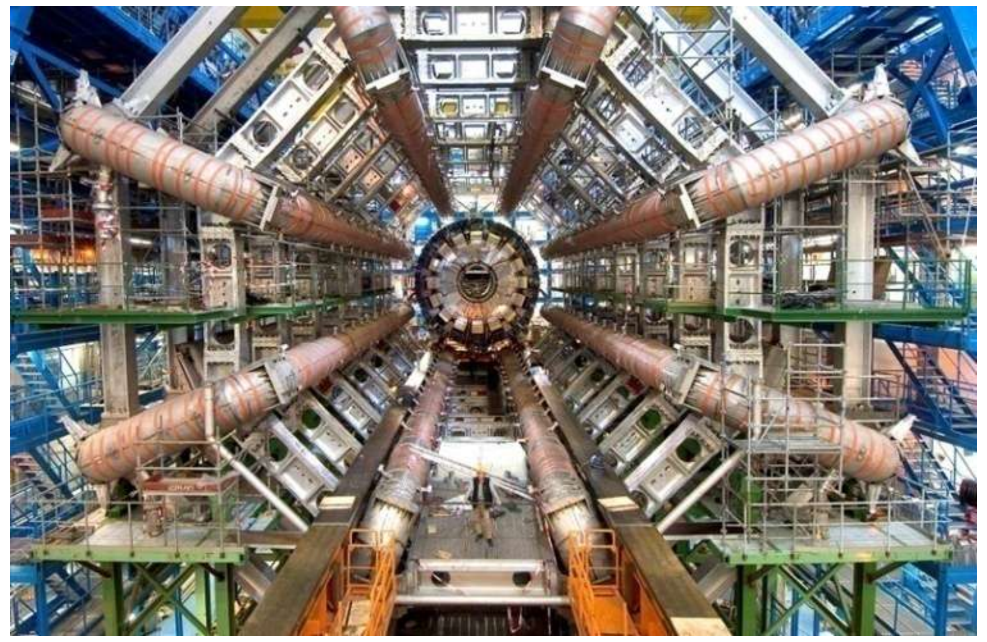
Quantum Field Theory Scattering Amplitudes

Scattering amplitudes give us quantum mechanical description of events at particle colliders.

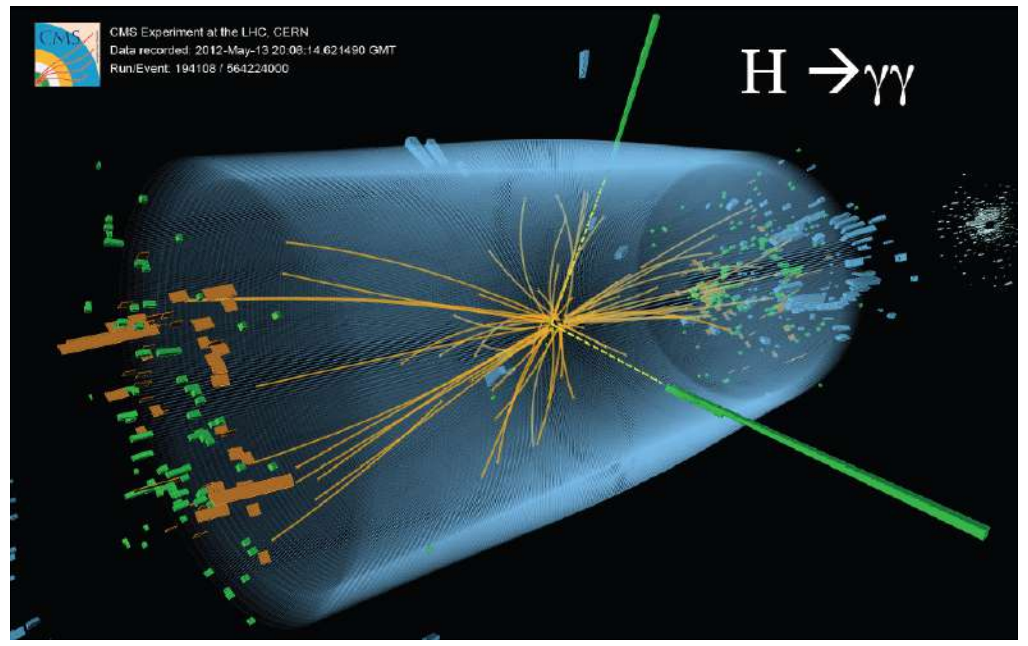


particle scattering

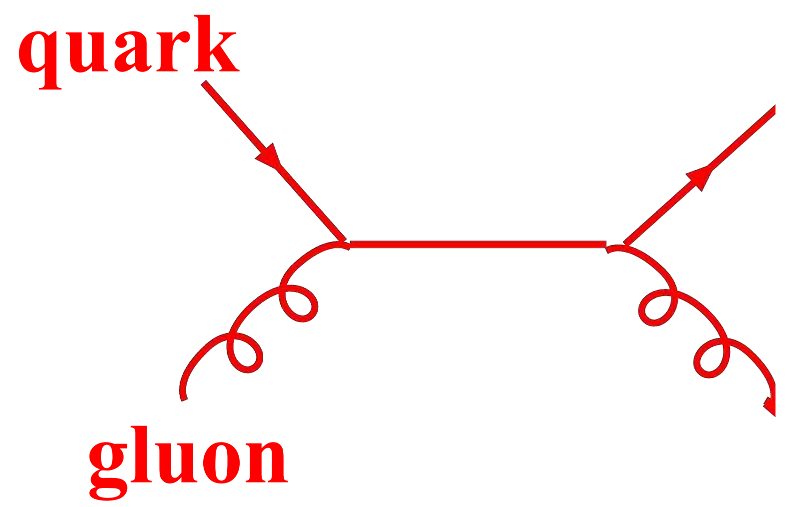
Large Hadron Collider



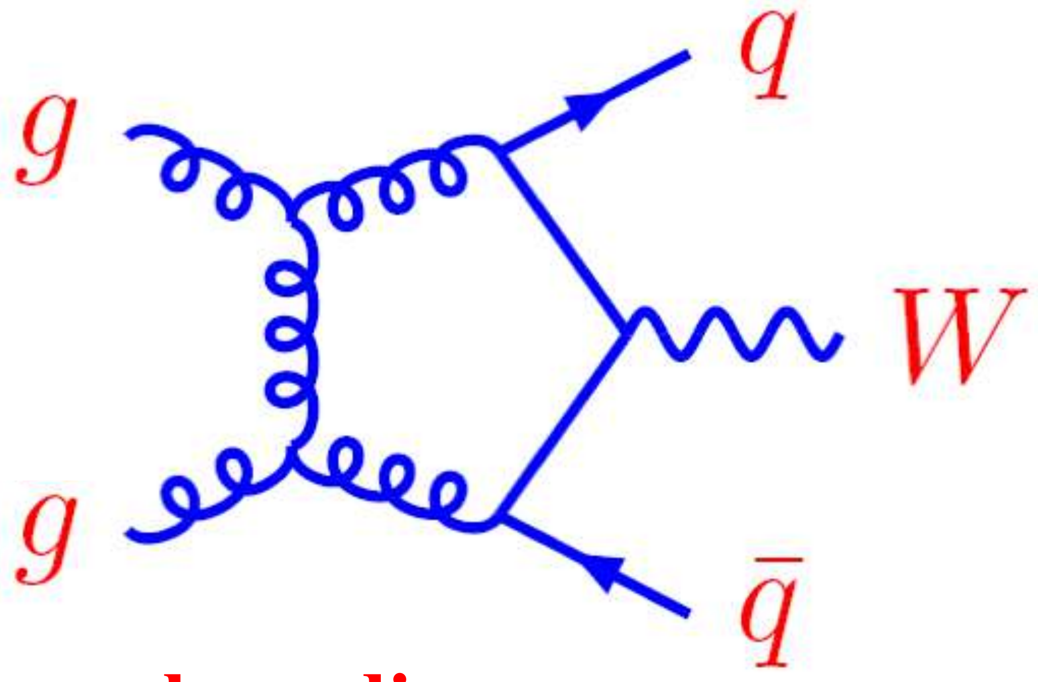
ATLAS Detector



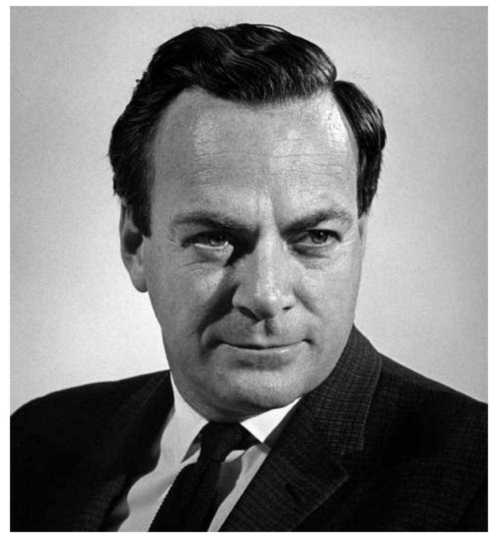
Higgs boson event



tree Feynman diagram



loop diagram
higher order



Scattering amplitudes provide a foundation for understanding particle collisions

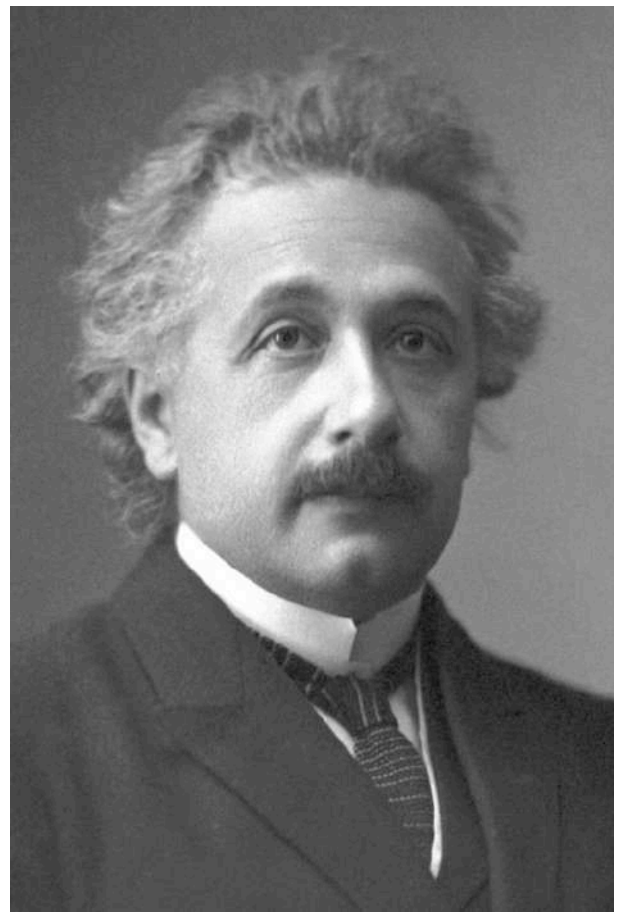
General Relativity

Usually, people start from Einstein's field equations:

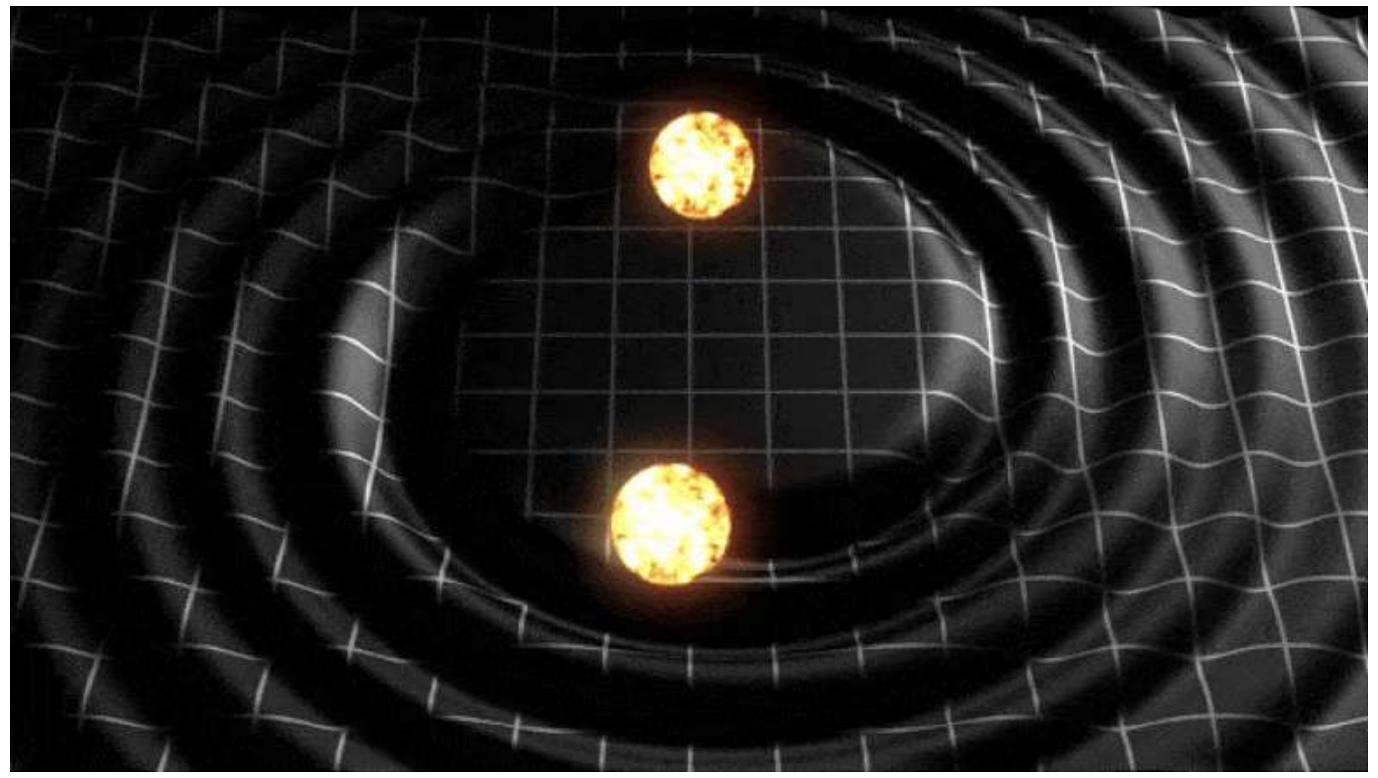
energy-momentum tensor

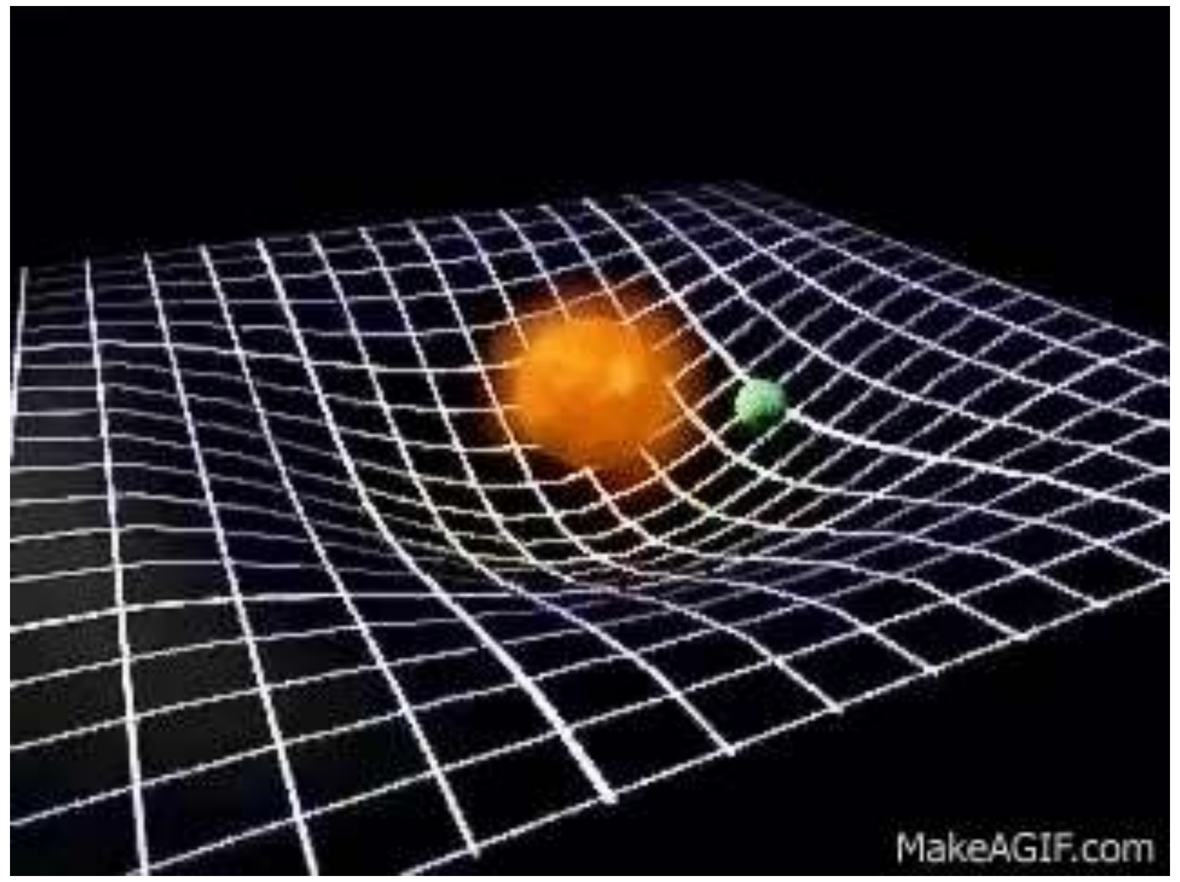
$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G T_{\mu\nu}$$

curvature



Geometry
Curvature of space time



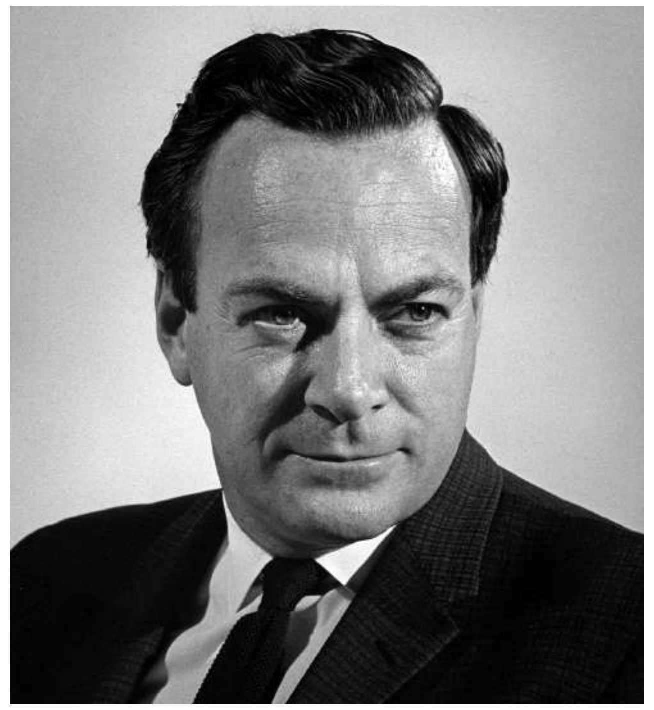


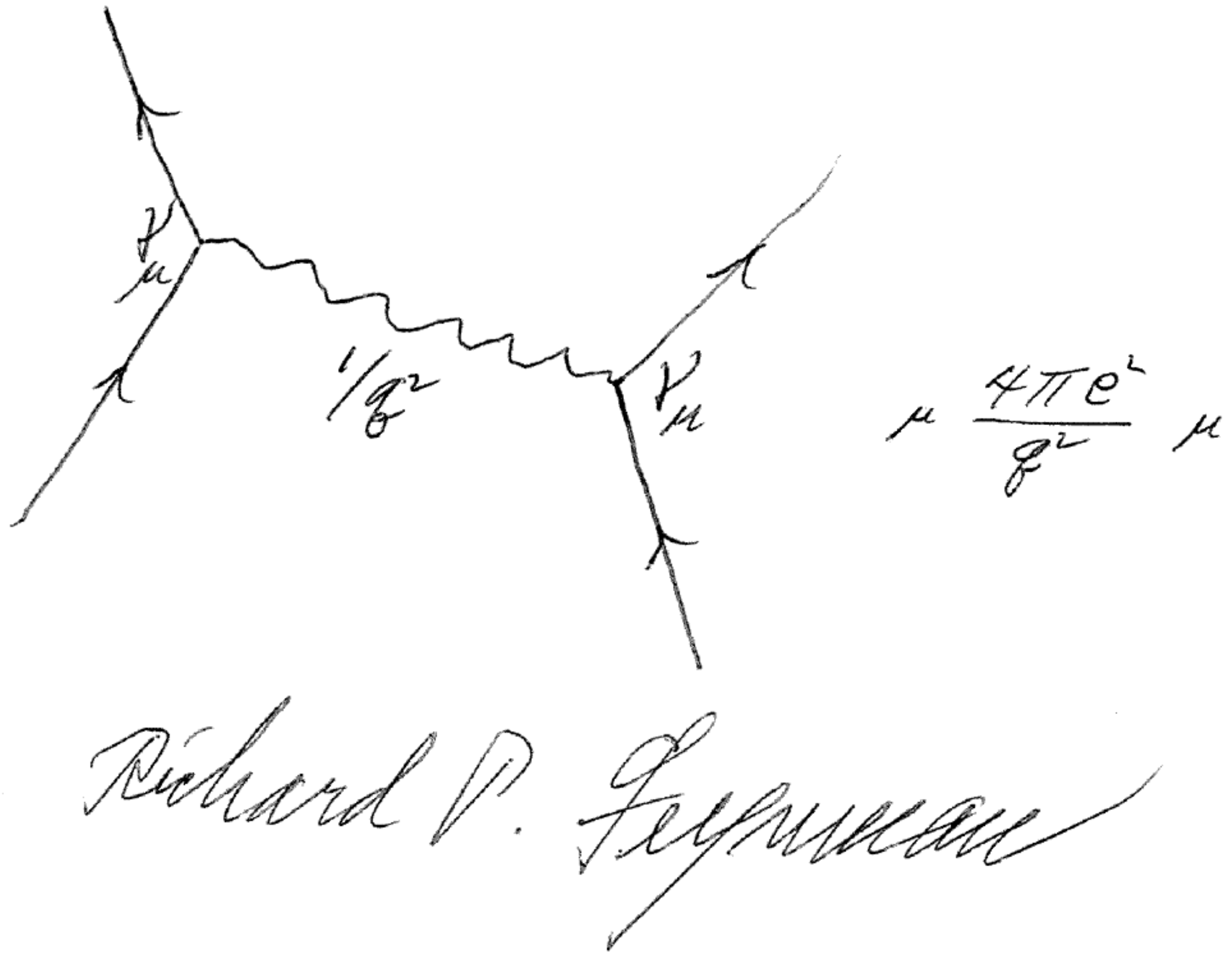
Amplitudes Approach to General Relativity

Amplitudes appoach does *not* start from Einstein's field equations.

$$\xcancel{R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}}$$

Feynman: Einstein's gravity follows if we assume existence of massless spin-2 particle





Our approach: Assume massless spin-2 gravitons exist

Are gravitons real? No one knows for sure (no measurements).

Whether real or not, we know for sure that they are useful!

Not suitable for all problems but very effective for certain problems.

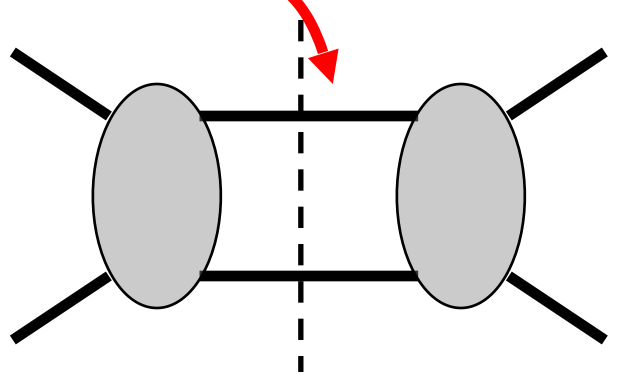
Basic Tool: Generalized Unitarity Method

ZB, Dixon, Dunbar and Kosower (1994)

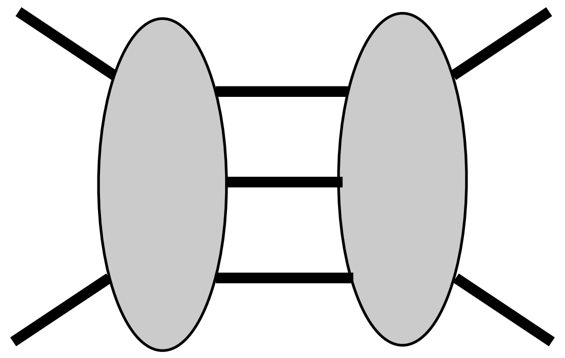
No Feynman rules; no need for virtual particles.

$E^2 = \vec{p}^{\,2} + m^2$ on-shell

Two-particle cut:

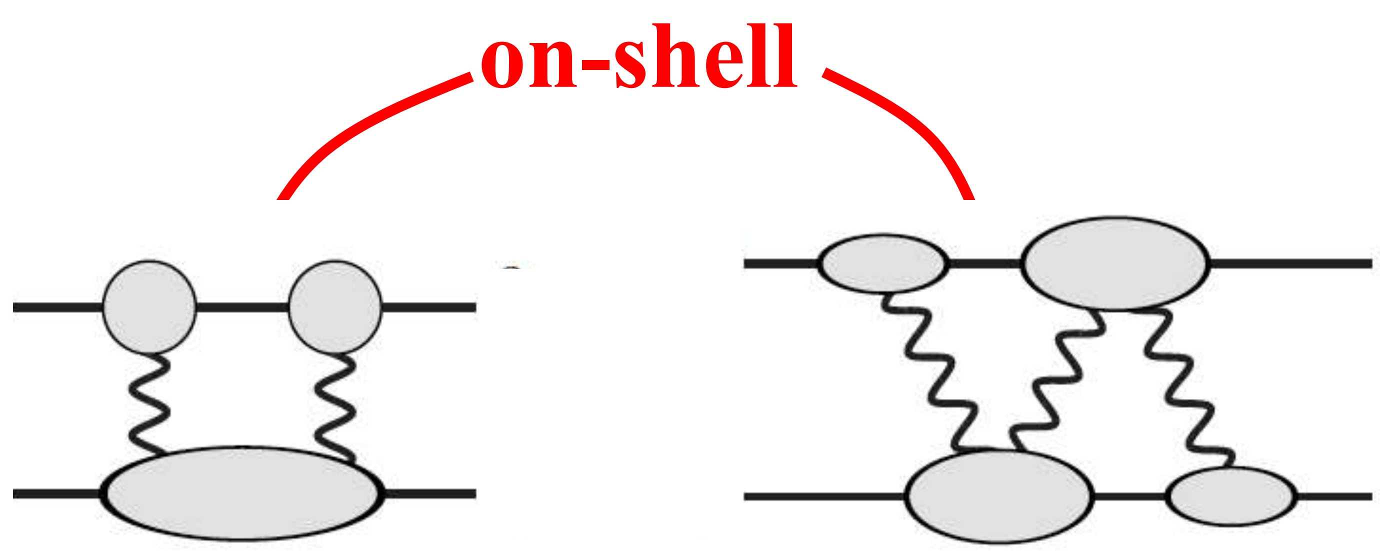


Three-particle cut:



- **Systematic assembly of complete loop amplitudes from tree amplitudes.**
- **Works for any number of particles or loops.**

Generalized unitarity as a practical tool for loops.



ZB, Dixon and Kosower;
ZB, Morgan;
Britto, Cachazo, Feng;
Ossala,Pittau,Papadopoulos;
Ellis, Kunszt, Melnikov;
Forde; Badger;
ZB, Carrasco, Johansson, Kosower
and many others

Designed to match results one would obtain from Feynman diagrams.
Idea used in the "NLO revolution" in QCD collider physics.
Here will discuss applications to gravity.

Gauge Theory vs Gravity

Consider the Einstein gravity Lagrangian

curvature

$$L_{\text{gravity}} = \frac{2}{\kappa^2}\sqrt{-g}\,R$$

$$\kappa^2 = 32\pi\, G_{\text{Newton}}$$

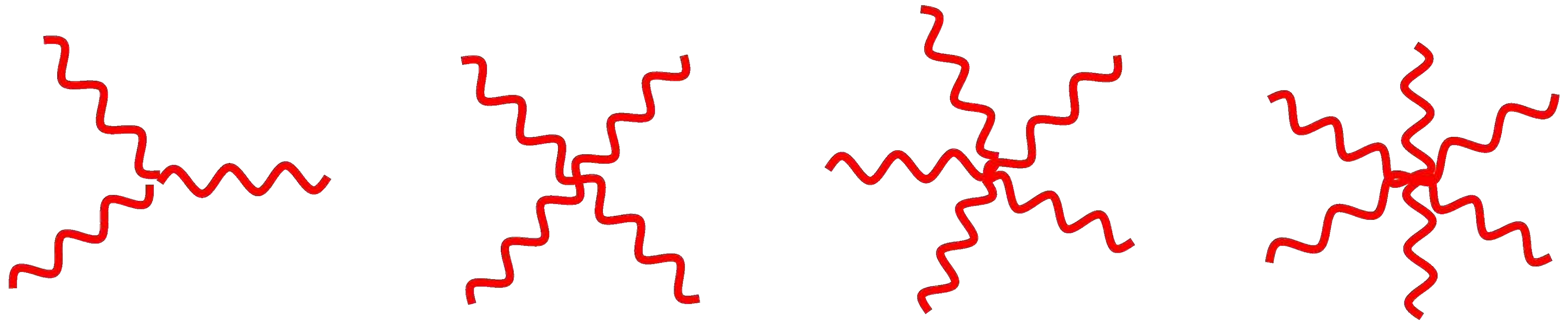
Flat-space metric

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$$

graviton field

metric

Infinite number of complicated interactions

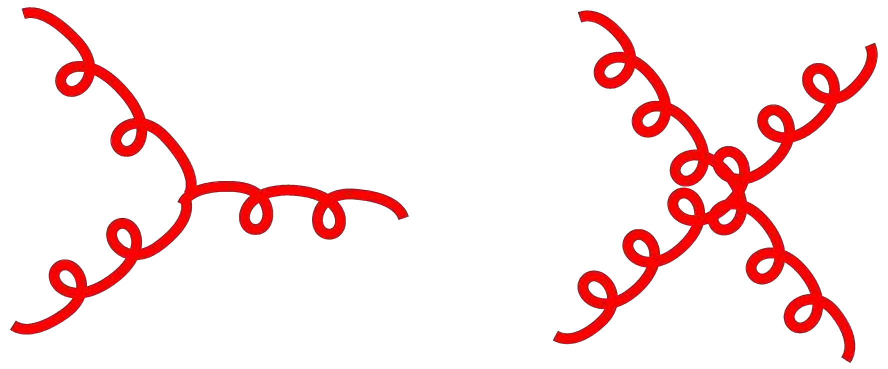


$+ \cdots$

Very nonlinear

Compare to gauge-theory Lagrangian on which QCD is based

$$L_{\text{YM}} = \frac{1}{g^2} F^2$$

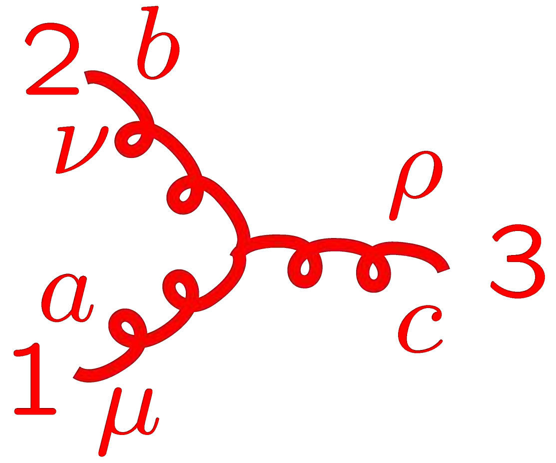


Only three and four point interactions

Gravity seems so much more complicated than gauge theory.

Three-Point Interactions

Standard perturbative approach:



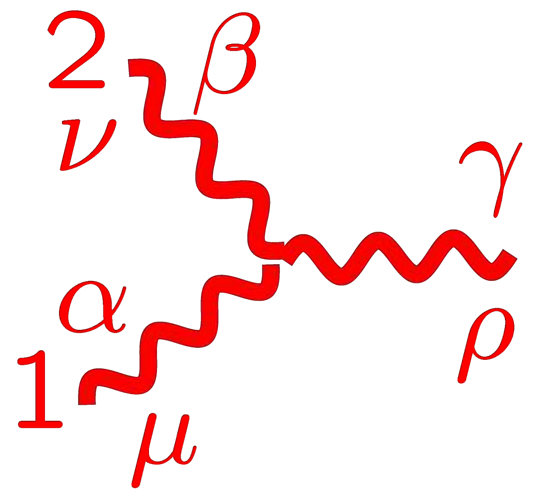
Three-gluon vertex from QCD strong interactions:

$$V_{3\,\mu\nu\sigma}^{abc} = -gf^{abc}(\eta_{\mu\nu}(k_1-k_2)_\rho+\eta_{\nu\rho}(k_1-k_2)_\mu+\eta_{\rho\mu}(k_1-k_2)_\nu)$$

Three-graviton vertex:

$$k_i^2 = E_i^2 - \vec{k}_i^2 \neq 0$$

$$\begin{aligned}
&G_{3\mu\alpha,\nu\beta,\sigma\gamma}(k_1,k_2,k_3) = \\
&\quad \mathrm{sym}[-\tfrac{1}{2}P_3(k_1\cdot k_2\eta_{\mu\alpha}\eta_{\nu\beta}\eta_{\sigma\gamma}) - \tfrac{1}{2}P_6(k_{1\nu}k_{1\beta}\eta_{\mu\alpha}\eta_{\sigma\gamma}) + \tfrac{1}{2}P_3(k_1\cdot k_2\eta_{\mu\nu}\eta_{\alpha\beta}\eta_{\sigma\gamma}) \\
&\qquad + P_6(k_1\cdot k_2\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\beta\gamma}) + 2P_3(k_{1\nu}k_{1\gamma}\eta_{\mu\alpha}\eta_{\beta\sigma}) - P_3(k_{1\beta}k_{2\mu}\eta_{\alpha\nu}\eta_{\sigma\gamma}) \\
&\qquad + P_3(k_{1\sigma}k_{2\gamma}\eta_{\mu\nu}\eta_{\alpha\beta}) + P_6(k_{1\sigma}k_{1\gamma}\eta_{\mu\nu}\eta_{\alpha\beta}) + 2P_6(k_{1\nu}k_{2\gamma}\eta_{\beta\mu}\eta_{\alpha\sigma}) \\
&\qquad + 2P_3(k_{1\nu}k_{2\mu}\eta_{\beta\sigma}\eta_{\gamma\alpha}) - 2P_3(k_1\cdot k_2\eta_{\alpha\nu}\eta_{\beta\sigma}\eta_{\gamma\mu})]
\end{aligned}$$



About 100 terms in three vertex

Naïve conclusion: Perturbative gravity is a mess.

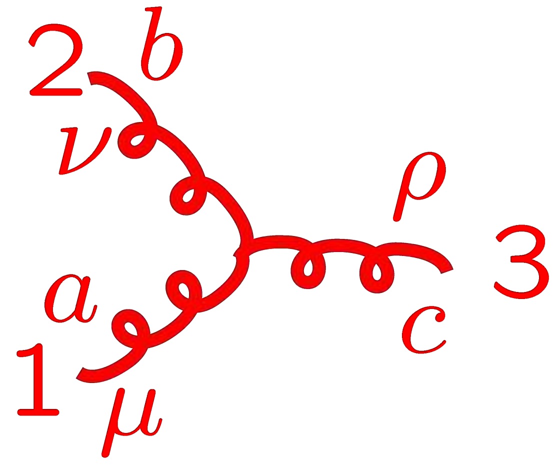
Simplicity of Gravity Amplitudes

People were looking at gravity amplitudes the wrong way.
On-shell viewpoint more powerful.

On-shell **three vertices contains all information:**

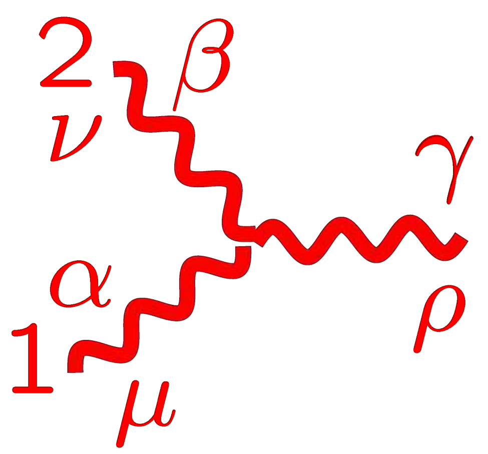
$$E_i^2 - \vec{k}_i^2 = 0$$

Yang-Mills gauge theory, QCD:



$$-g f^{abc}(\eta_{\mu\nu}(k_1 - k_2)_\rho + \text{cyclic})$$

Einstein gravity:



$$i\kappa(\eta_{\mu\nu}(k_1 - k_2)_\rho + \text{cyclic}) \times (\eta_{\alpha\beta}(k_1 - k_2)_\gamma + \text{cyclic})$$

"square" of Yang-Mills vertex.

Simple interactions dictated by gravitons being massless spin 2 particles.

KLT Relation Between Gravity and Gauge Theory

KLT (1985)

Kawai-Lewellen-Tye string relations in low-energy limit:

gravity

gauge-theory color ordered

$$M_4^{\rm tree}(1,2,3,4) = -is_{12} A_4^{\rm tree}(1,2,3,4)\, A_4^{\rm tree}(1,2,4,3)\,,$$

$$M_5^{\rm tree}(1,2,3,4,5) = is_{12}s_{34} A_5^{\rm tree}(1,2,3,4,5)\, A_5^{\rm tree}(2,1,4,3,5) + is_{13}s_{24} A_5^{\rm tree}(1,3,2,4,5)\, A_5^{\rm tree}(3,1,4,2,5)$$

Generalizes to explicit all-leg form. ZB, Dixon, Perelstein, Rozowsky

1. **Gravity is derivable from gauge theory. Standard Lagrangian methods offers no hint why this is possible.**

2. **It is very generally applicable.**

Duality Between Color and Kinematics

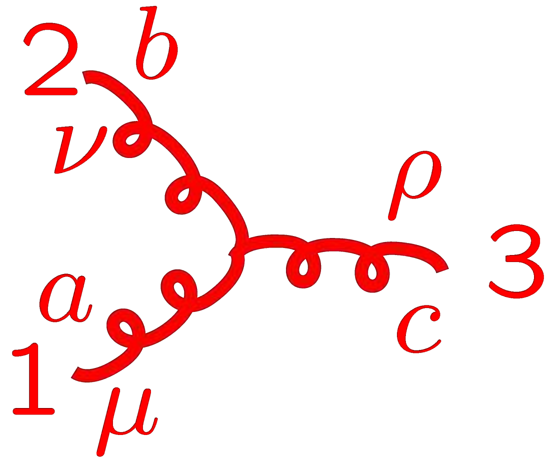
ZB, Carrasco, Johansson (BCJ)

coupling constant

color factor

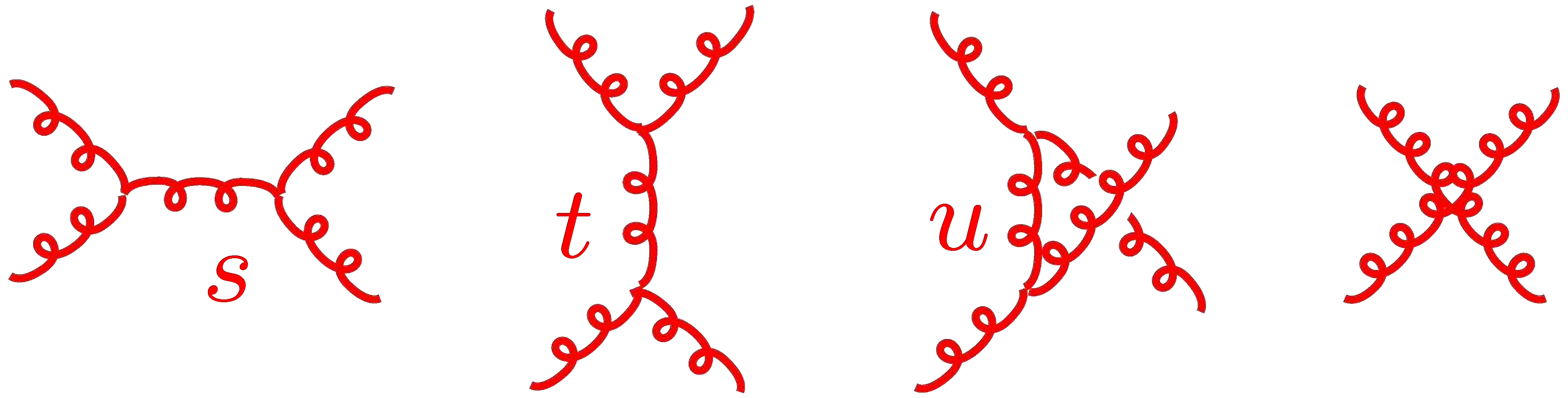
momentum dependent kinematic factor

$$-gf^{abc}(\eta_{\mu\nu}(k_1 - k_2)_\rho + \text{cyclic})$$



Color factors based on a Lie algebra: $[T^a, T^b] = if^{abc}T^c$

Jacobi Identity $f^{a_1a_2b}f^{ba_4a_3} + f^{a_4a_2b}f^{ba_3a_1} + f^{a_4a_1b}f^{ba_2a_3} = 0$



Use 1 = *s/s* = *t/t* = *u/u* to assign 4-point diagram to others.

$$s = (k_1 + k_2)^2 \quad t = (k_1 + k_4)^2$$
$$u = (k_1 + k_3)^2$$

$$\mathcal{A}_4^{\text{tree}} = g^2\left(\frac{n_s c_s}{s} + \frac{n_t c_t}{t} + \frac{n_u c_u}{u}\right)$$

Color factors satisfy Jacobi identity:

Numerator factors satisfy similar identity:

$$\boxed{\begin{aligned} c_u &= c_s - c_t \\ n_u &= n_s - n_t \end{aligned}}$$

The secret of perturbative gravity is found here.

Gravity as a Double Copy of Gauge Theory

ZB, Carrasco, Johansson

gauge theory (QCD):

$$\mathcal{A}_n^{\text{tree}} = i g^{n-2} \sum_i \frac{c_i\, n_i}{D_i}$$

color factor

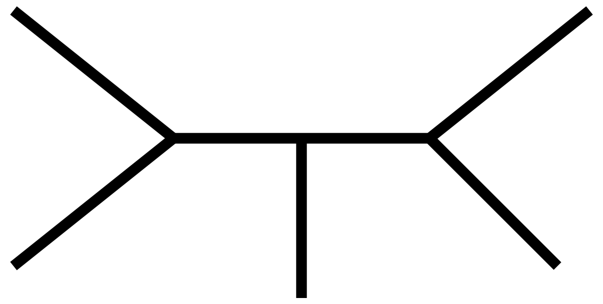
kinematic numerator factor

Feynman propagators

Jacobi identities

$$c_k = c_i - c_j$$
$$n_k = n_i - n_j$$

$$\boxed{c_i \to n_i}$$



sum over diagrams with only 3 vertices

Einstein gravity:

$$\mathcal{M}_n^{\text{tree}} = i \kappa^{n-2} \sum_i \frac{n_i^2}{D_i}$$

$$n_i \sim k_4 \cdot k_5\, k_2 \cdot \varepsilon_1\, \varepsilon_2 \cdot \varepsilon_3\, \varepsilon_4 \cdot \varepsilon_5 + \cdots$$

Gravity and gauge theory kinematic numerators are the same!

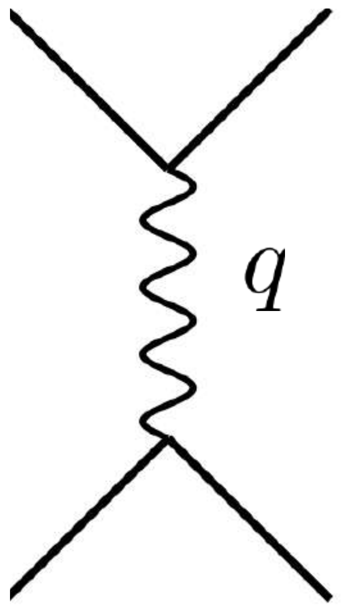
Cries out for a unified description of gravity with gauge theory, presumably along the lines of string theory.

2 Body Potentials and Amplitudes

Tree-level: Fourier transform gives classical potential.

$$V(r) \sim \int \frac{d^3q}{(2\pi)^3} e^{-i\mathbf{q}\cdot\mathbf{r}} A^{\text{tree}}(\mathbf{q})$$

$$H(\mathbf{p},\mathbf{r}) = \sqrt{\mathbf{p}^2 + m_1^2} + \sqrt{\mathbf{p}^2 + m_2^2} + V(\mathbf{p},\mathbf{r})$$



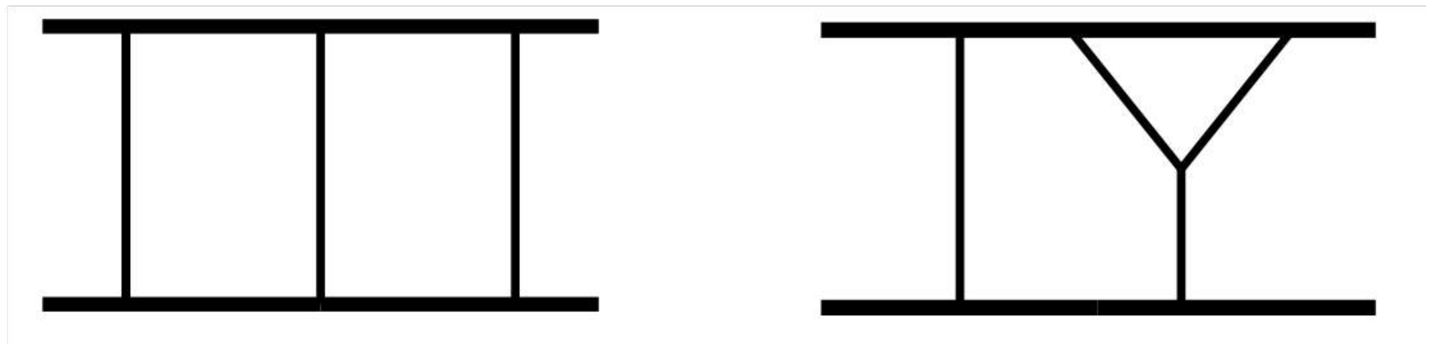
Newtonian potential follows follows from tree amplitude

$$V(\mathbf{p},\mathbf{r}) = \sum_{i=1}^{\infty} c_i(\mathbf{p}^2)\left(\frac{G_N}{|\mathbf{r}|}\right)^i$$

Beyond 1 loop more complicated:

- **Loops have classical pieces.**
- $1/\hbar^L$ **scaling at** L **loop.**
- **Double counting and iteration.**

$$e^{iS_{\text{classical}}/\hbar}$$



Piece of loops are classical: Our task is to efficiently extract these pieces. Also radiative and dissipative effects are contained in the amplitude.

General Relativity: Unitarity + Double Copy

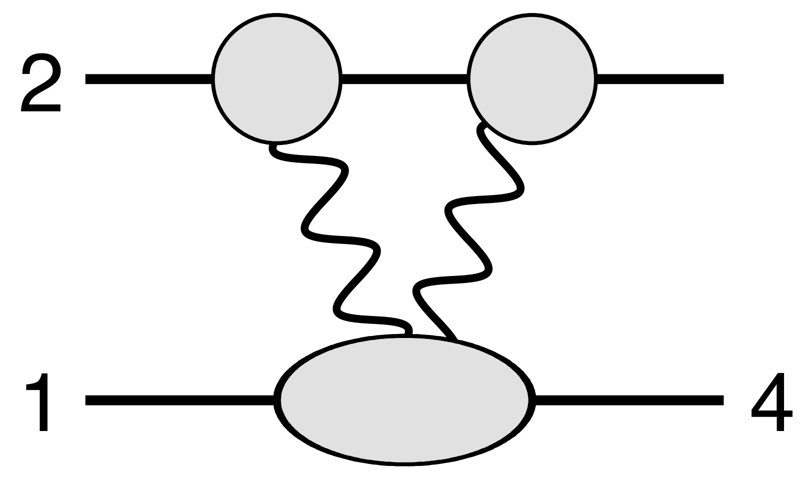
- Long-range force: Two matter lines separated by graviton propagators.
- Classical physics: 1 matter line per loop is cut (on-shell).

$$E^2 = \vec{p}^{\,2} + m^2$$

Only independent unitarity cut for $O(G^2)$.

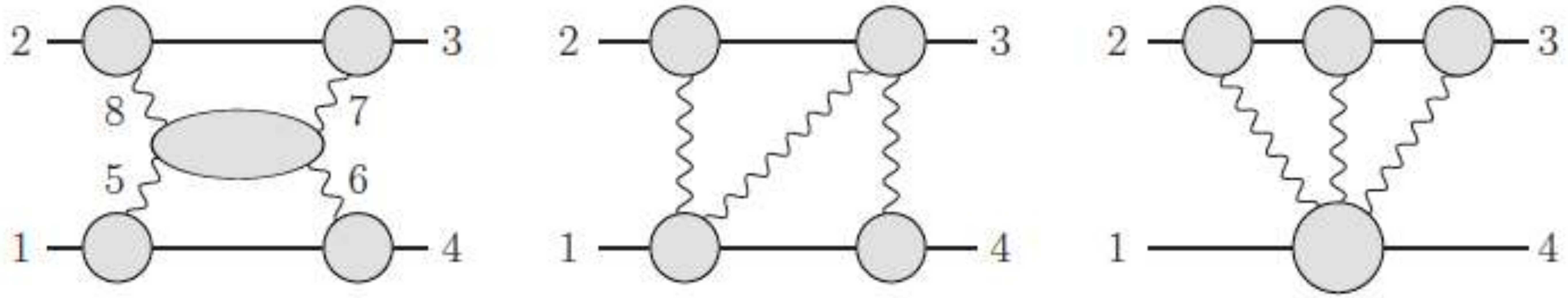
G is Newton's constant

2 black hole scattering

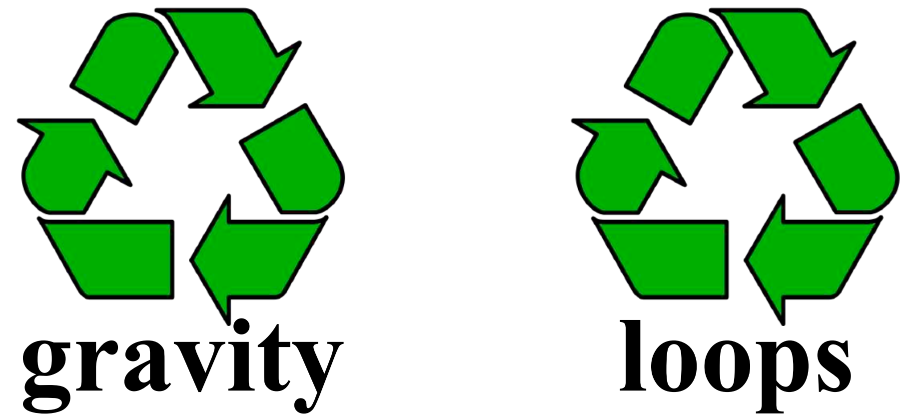


Treat exposed lines on-shell (long range).
Pieces we want are simple!

Independent generalized unitarity cuts for $O(G^3)$.



Amplitude tools fit well for extracting the pieces we want.



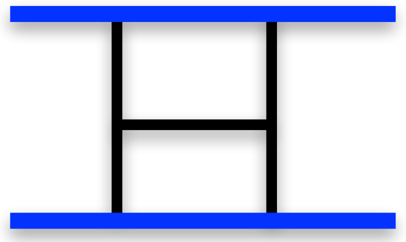
Improved Methods for Extracting Classical Physics.

There are now multiple alternative ways to extract classical physics.

- **EFT matching to 2 body Hamiltonian** — Cheng, Solon, Rothstein; ZB, Cheung, Roiban, Shen, Zeng
- **Map to EOB** — Bini, Damour, Geralico
- **Calculate physical observables** — Kosower, Maybee, O'Connell (KMOC)
- **Eikonal phase** — Amati, Ciafaloni, Veneziano; Di Vecchia, Heissenberg, Russo, Veneziano
- **Amplitude radial-action relation** — ZB, Parra-Martinez, Roiban, Ruf, Shen, Solon, Zeng
- **Exponential representation** — Damgaard, Plante, Vanhove; Bjerrum-Bohr, Plante, Vanhove
- **Heavy mass field theory** — Brandhuber, Chen, Travaglini, Wen; Damgaard, Haddad, Helset
- **Worldline formalisms** — Goldberger, Rothstein; Levi, Steinhoff; Dlapa, Kälin, Liu, Porto; Jakobson, Mogul, Plefka, Steinhoff; Edison, Levi; etc
- **Magnusian approach** — Kim, Patil, Sheopner, Steinhoff; Guo, J. Kim, J. Kim, S. Kim, Lee, Li; Gonzo, Mogull

These days, radial action and KMOC are our preferred methods.

Amplitude in Conservative Classical Potential Limit



ZB, Cheung, Roiban, Shen, Solon, Zeng (2019)

First paper beyond 2PM: The O(G^3) or 3PM conservative terms are:

$$\mathcal{M}_3 = \frac{\pi G^3 \nu^2 m^4 \log \boldsymbol{q}^2}{6\gamma^2 \xi}\Bigg[3 - 6\nu + 206\nu\sigma - 54\sigma^2 + 108\nu\sigma^2 + 4\nu\sigma^3 - \frac{48\nu\left(3+12\sigma^2-4\sigma^4\right)\operatorname{arcsinh}\sqrt{\frac{\sigma-1}{2}}}{\sqrt{\sigma^2-1}} - \frac{18\nu\gamma\left(1-2\sigma^2\right)\left(1-5\sigma^2\right)}{(1+\gamma)(1+\sigma)}\Bigg] + \frac{8\pi^3 G^3 \nu^4 m^6}{\gamma^4 \xi}\Big[3\gamma\left(1-2\sigma^2\right)\left(1-5\sigma^2\right)F_1 - 32 m^2\nu^2\left(1-2\sigma^2\right)^3 F_2\Big]$$

$$m = m_A + m_B, \qquad \mu = m_A m_B/m, \qquad \nu = \mu/m, \qquad \gamma = E/m,$$

$$\xi = E_1 E_2/E^2, \qquad E = E_1 + E_2, \qquad \sigma = p_1 \cdot p_2/m_1 m_2,$$

- **Amplitude remarkably compact.**
- **Arcsinh and the appearance of a mass singularity is new feature.** $\log(E^2/m_1 m_2)$

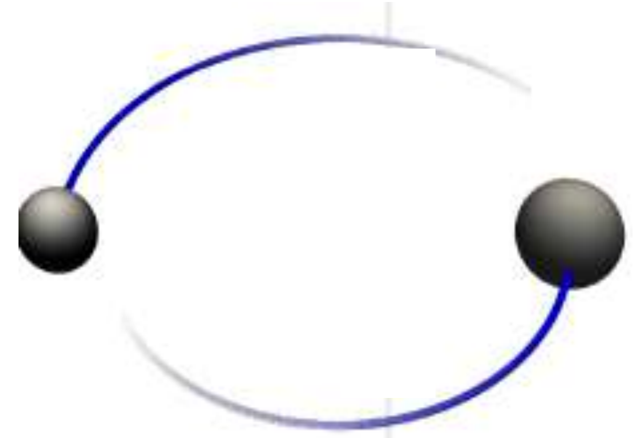
Di Vecchia, Heissenberg, Russo, Veneziano; Damour

- **Derived conservative scattering angle has simple mass dependence.**

Antonelli, Buonanno, Steinhoff, van de Meent, Vines
Comprehensive understanding: Damour

Our First Paper: Conservative 3PM Hamiltonian

ZB, Cheung, Roiban, Shen, Solon, Zeng (2019)



The 3PM two body Hamiltonian:

$$H(\boldsymbol{p},\boldsymbol{r}) = \sqrt{\boldsymbol{p}^2+m_1^2}+\sqrt{\boldsymbol{p}^2+m_2^2}+V(\boldsymbol{p},\boldsymbol{r})$$

$$V(\boldsymbol{p},\boldsymbol{r}) = \sum_{i=1}^{3} c_i(\boldsymbol{p}^2)\left(\frac{G}{|\boldsymbol{r}|}\right)^i,$$

Newton in here

$$c_1 = \frac{\nu^2 m^2}{\gamma^2\xi}\left(1-2\sigma^2\right), \qquad c_2 = \frac{\nu^2 m^3}{\gamma^2\xi}\left[\frac{3}{4}\left(1-5\sigma^2\right)-\frac{4\nu\sigma\left(1-2\sigma^2\right)}{\gamma\xi}-\frac{\nu^2(1-\xi)\left(1-2\sigma^2\right)^2}{2\gamma^3\xi^2}\right],$$

$$c_3 = \frac{\nu^2 m^4}{\gamma^2\xi}\left[\frac{1}{12}\left(3-6\nu+206\nu\sigma-54\sigma^2+108\nu\sigma^2+4\nu\sigma^3\right)-\frac{4\nu\left(3+12\sigma^2-4\sigma^4\right)\operatorname{arcsinh}\sqrt{\frac{\sigma-1}{2}}}{\sqrt{\sigma^2-1}}\right.$$
$$-\frac{3\nu\gamma\left(1-2\sigma^2\right)\left(1-5\sigma^2\right)}{2(1+\gamma)(1+\sigma)}-\frac{3\nu\sigma\left(7-20\sigma^2\right)}{2\gamma\xi}-\frac{\nu^2\left(3+8\gamma-3\xi-15\sigma^2-80\gamma\sigma^2+15\xi\sigma^2\right)\left(1-2\sigma^2\right)}{4\gamma^3\xi^2}$$
$$\left.+\frac{2\nu^3(3-4\xi)\sigma\left(1-2\sigma^2\right)^2}{\gamma^4\xi^3}+\frac{\nu^4(1-2\xi)\left(1-2\sigma^2\right)^3}{2\gamma^6\xi^4}\right],$$

$$m = m_A+m_B, \qquad \mu = m_A m_B/m, \qquad \nu = \mu/m, \qquad \gamma = E/m,$$
$$\xi = E_1E_2/E^2, \qquad E = E_1+E_2, \qquad \sigma = p_1\cdot p_2/m_1m_2,$$

Scattering amplitude methods are useful for gravitational-wave problem

How do we know it is right?

Original checks:

- Compared to 4PN Hamiltonians (after canonical transformation)
- In test mass limit, $m_1 \ll m_2$, matches Schwarzschild Hamiltonian

Damour, Jaranowski, Schäfer; Bernard, Blanchet, Bohé, Faye, Marsat

Results met with initial skepticism. Damour, arXiv:1912.02139v1

Subsequent calculations confirm our 3PM result:

1. Subsequent papers confirm our result in 6PN overlap after initial skepicism.

   Blümlein, Maier, Marquard, Schäfer; Bini, Damour, Geralico

2. New calculations reproducing 3PM result.

   Cheung and Solon; Kälin, Liu, Porto;

3. Adding (non-conservative) real radiation clarifies a puzzle at high energies.

   Di Vecchia, Heissenberg, Russo, Veneziano; Damour

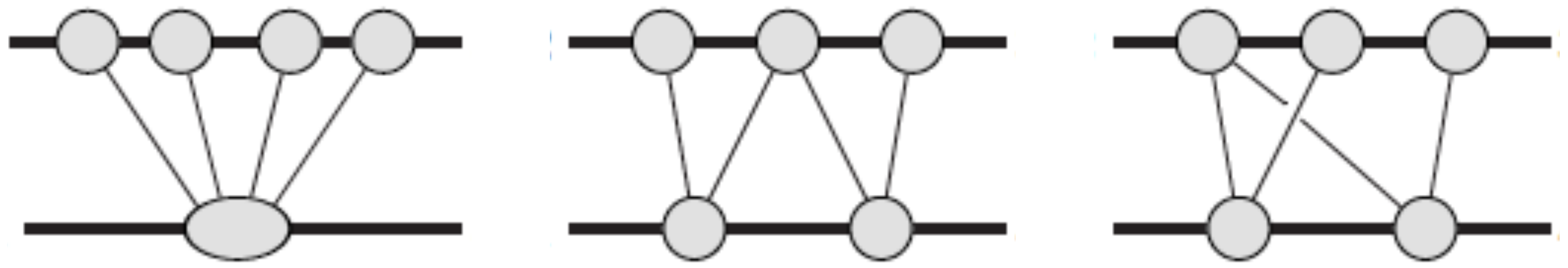
3PM results have passed highly nontrivial checks and careful scrutiny.

Higher Orders: O(G^4) & O(G^5)

Methods scale well to higher orders

ZB, Parra-Martinez, Roiban, Ruf, Shen, Solon, Zeng (2022)

O(G^4)

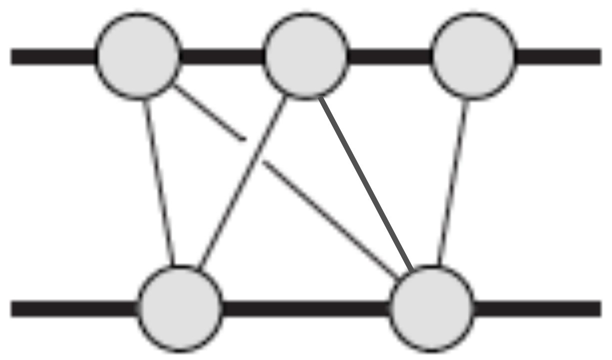


etc

***O(G^4)* done,**
but nontrivial time nonlocality issues not completely resolved.

Dissipative effects: Manohar, Shen and Ridgeway; Dlapa, Kalen, Lui, Neef, Porto; Damgaard, Hansen, Planté, Vanhove.

O(G^5)

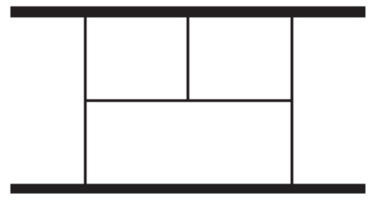


Good progress. Integrals are the main issue.

ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng.
Klemm, Naga, Sauer, Plefka; Driesse, Jakobson, Mogull, Plefka, Sauer, Usovitch.

Goal is O(G^6): Need factor 100 precision increase. Highly Nontrivial!

Complete Conservative Contribution O(G^4)



ZB, Parra-Martinez, Roiban, Ruf, Shen, Solon, Zeng

O(G^4) amplitude

test particle

1st self force

Iteration. No need to compute

$$\mathcal{M}_4^{\rm cons} = G^4 M^7 \nu^2 |\boldsymbol{q}| \pi^2 \Big[\mathcal{M}_4^{\rm p} + \nu\Big(4\mathcal{M}_4^{\rm t}\log\big(\tfrac{p_\infty}{2}\big) + \mathcal{M}_4^{\pi^2} + \mathcal{M}_4^{\rm rem}\Big)\Big] + \int_\ell \frac{\tilde I_{r,1}^4}{Z_1 Z_2 Z_3} + \int_\ell \frac{\tilde I_{r,1}^2 \tilde I_{r,2}}{Z_1 Z_2} + \int_\ell \frac{\tilde I_{r,1} \tilde I_{r,3}}{Z_1} + \int_\ell \frac{\tilde I_{r,2}^2}{Z_1}$$

tail effect

$$D = 4 - 2\epsilon$$

$$\mathcal{M}_4^{\rm t} = r_1 + r_2 \log\left(\tfrac{\sigma+1}{2}\right) + r_3 \frac{\operatorname{arccosh}(\sigma)}{\sqrt{\sigma^2-1}}\,, \qquad \mathcal{M}_4^{\rm p} = -\frac{35\left(1 - 18\sigma^2 + 33\sigma^4\right)}{8\left(\sigma^2-1\right)}$$

$$\mathcal{M}_4^{\pi^2} = r_4 \pi^2 + r_5\, \mathrm{K}\left(\tfrac{\sigma-1}{\sigma+1}\right)\mathrm{E}\left(\tfrac{\sigma-1}{\sigma+1}\right) + r_6\, \mathrm{K}^2\left(\tfrac{\sigma-1}{\sigma+1}\right) + r_7\, \mathrm{E}^2\left(\tfrac{\sigma-1}{\sigma+1}\right),$$

elliptic integrals

$$\mathcal{M}_4^{\rm rem} = r_8 + r_9 \log\left(\tfrac{\sigma+1}{2}\right) + r_{10} \tfrac{\operatorname{arccosh}(\sigma)}{\sqrt{\sigma^2-1}} + r_{11}\log(\sigma) + r_{12}\log^2\left(\tfrac{\sigma+1}{2}\right) + r_{13}\tfrac{\operatorname{arccosh}(\sigma)}{\sqrt{\sigma^2-1}}\log\left(\tfrac{\sigma+1}{2}\right) + r_{14}\tfrac{\operatorname{arccosh}^2(\sigma)}{\sigma^2-1}$$
$$+ r_{15}\,\mathrm{Li}_2\left(\tfrac{1-\sigma}{2}\right) + r_{16}\,\mathrm{Li}_2\left(\tfrac{1-\sigma}{1+\sigma}\right) + r_{17}\tfrac{1}{\sqrt{\sigma^2-1}}\left[\mathrm{Li}_2\left(-\sqrt{\tfrac{\sigma-1}{\sigma+1}}\right) - \mathrm{Li}_2\left(\sqrt{\tfrac{\sigma-1}{\sigma+1}}\right)\right].$$

$\nu = m_1 m_2/(m_1+m_2)^2$ $\qquad$ $\sigma = p_1 \cdot p_2 / m_1 m_2,$ $\qquad$ r_i rational coefficients

This is *complete* conservative contribution.

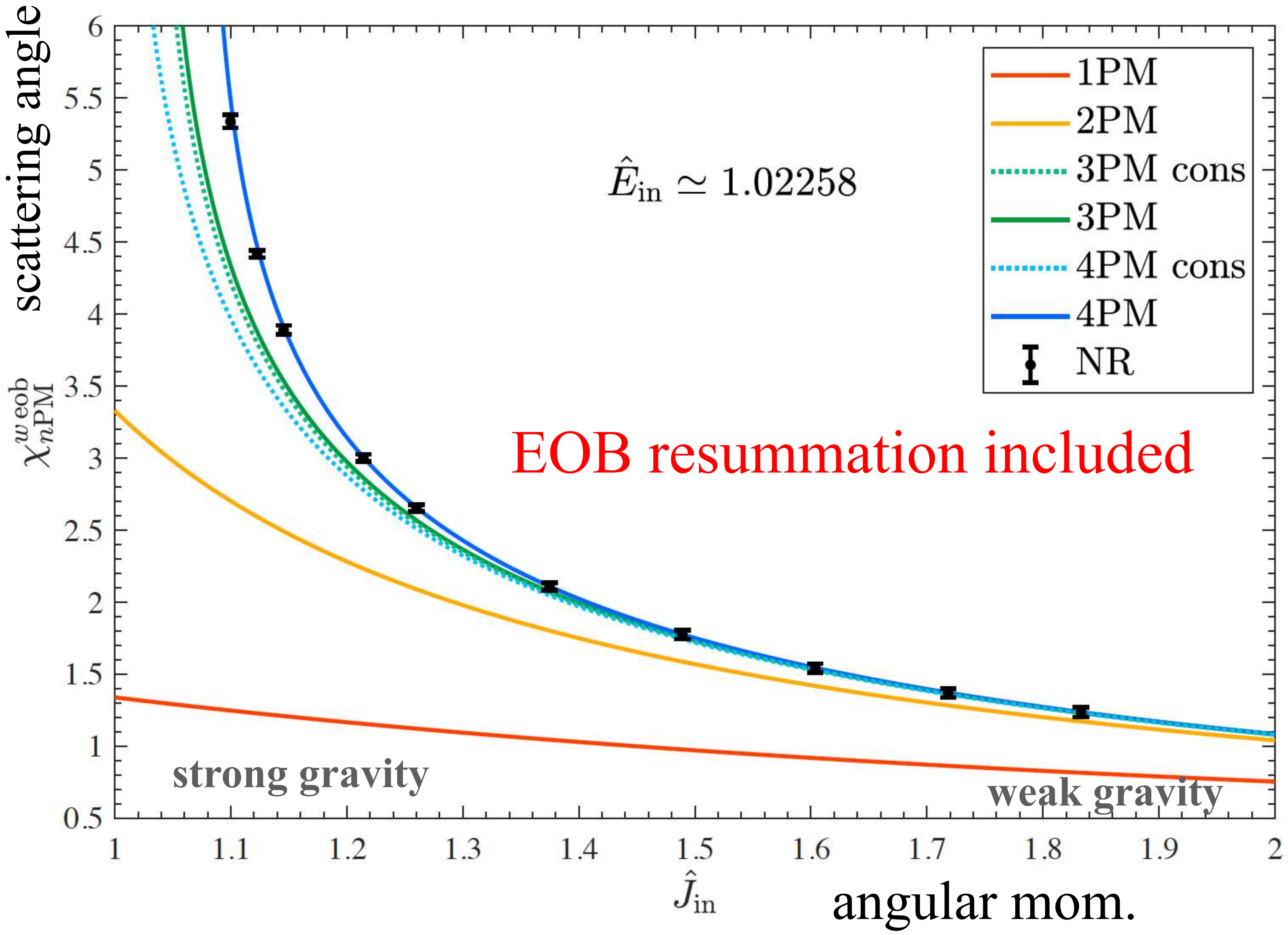
$$\mathcal{M}_4^{\rm radgrav,f} = \frac{12044}{75}p_\infty^2 + \frac{212077}{3675}p_\infty^4 + \frac{115917979}{793800}p_\infty^6 - \frac{9823091209}{76839840}p_\infty^8 + \frac{115240251793703}{1038874636800}p_\infty^{10} + \cdots$$

First 3 terms match 6PN results of Bini, Damour, Geralico!

Nontrivial analytic continuation of certain time nonlocal terms.
Partial patch. Khalil, Buonanno, Steinhoff, Vines; Dlapa, Kalin, Liu, Porto; Damour, Rettegno

Comparison with Numerical Relativity in Overlap Region

Khalil, Buonanno, Vines, Steinhoff; Damour and Rettegno



Plots use:

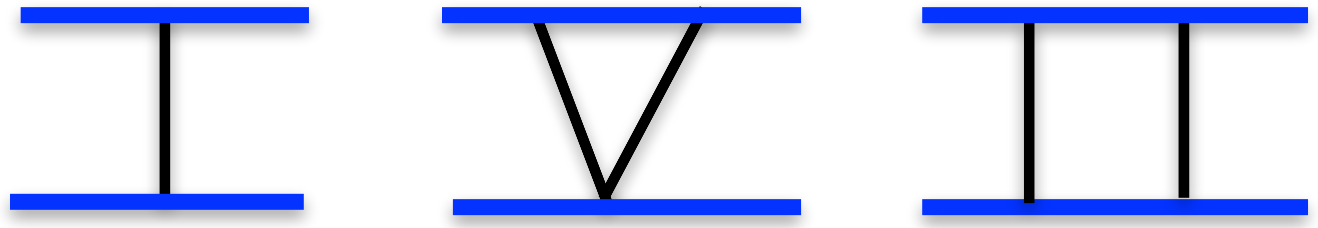
4PM Conservative: ZB, Parra-Martinez, Roiban, Ruf, Shen, Solon, Zeng
4PM Dissipative: Manohar, Shen and Ridgeway; Dlapa, Kalen, Lui, Neef, Porto
NR: Damour, Guercilena, Hinder, Hopper, Nagar, Rezzolla

- **Strikingly good agreement with numerical relativity.**
- **Shows we are on a good track!**

New Challenges and Theoretical Issues at Each Order

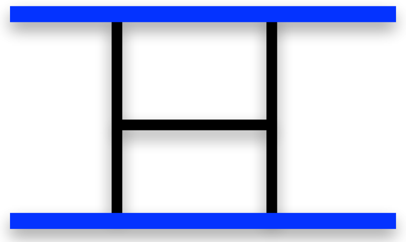
Moving up in orders of PM new effects and features encountered:

1PM and 2PM: **Fixed by geodesic motion, 0SF.**



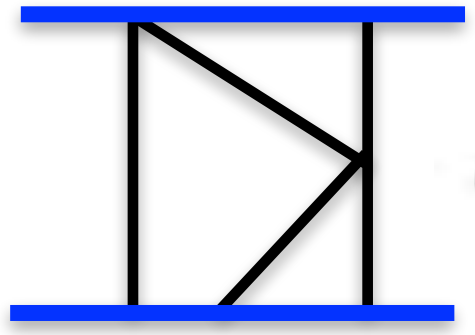
$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \frac{1}{1 - \frac{2M}{r}} dr^2 + r^2 d\theta^2 + r^2 \sin^2(\theta)\, d\phi^2$$

3PM: **Interesting structure in high energy limit. 1SF, m_1/m_2, beyond geodesic motion**



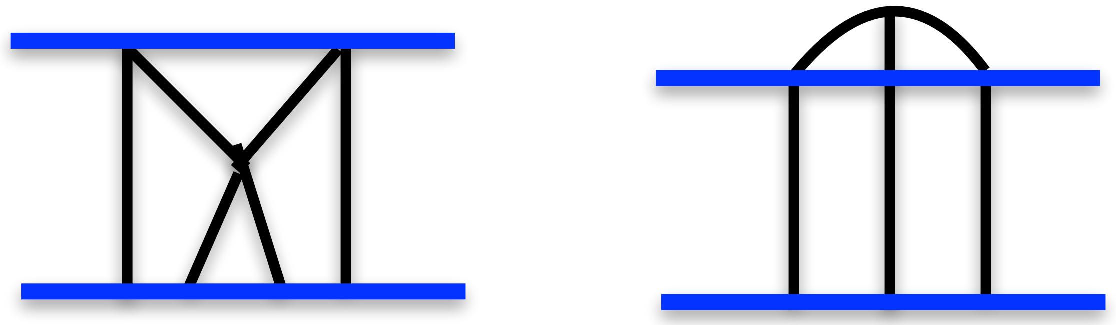
$\log(E^2/m_1 m_2)$ Cancels against real radiation

4PM: **Tail effect, nontrivial analytic continuations, elliptic integrals, *non*cancellation of poor high-energy behavior. Nonlocal in time "tail" effect.**

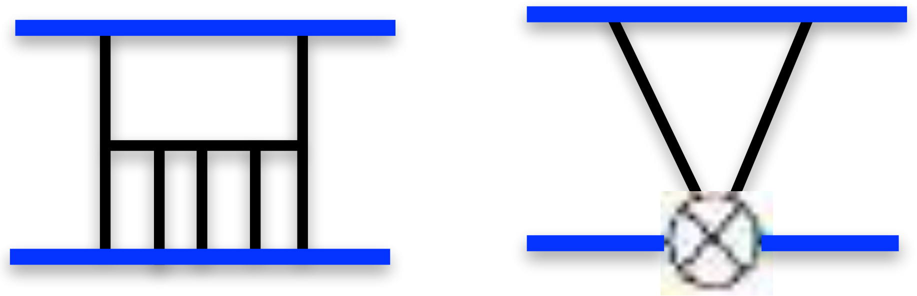


$$\sim K^2\left(\frac{\sigma-1}{\sigma+1}\right)$$

5PM: **2SF, new nontrivial functions.**



6PM: **Mixing with tidal operators, UV divergences.**
Distinguish BHs from neutron stars.
Renormalization in classical physics.



Towards 5PM, O(G^5)

ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov; Zeng;
Driesse, Jakobsen, Klemm, Mogull, Nega Plefka, Sauer, Usovitsch

Scattering Amplitudes

Double copy
Generalized unitarity
Expansion in classical limit

↓ **Straightforward (done)**

Integrand

Reduction to master integrals
DE's for master integrals
Solutions of DEs.

↓ **Hard**

Integrals are hard.

FIRE6 → FIRE7 → private code

Feedback to collider physics

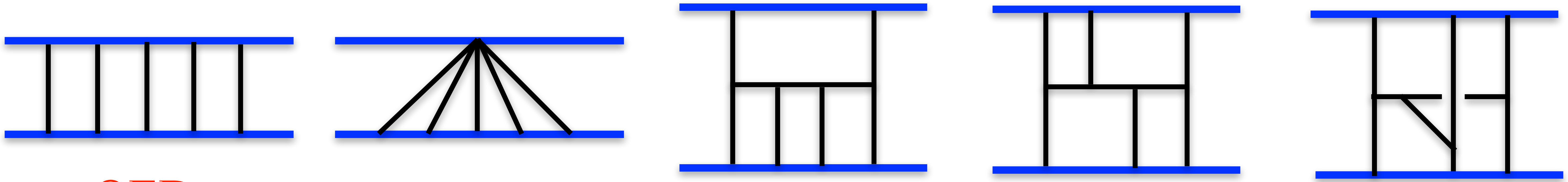
Integrated Amplitude

Eikonal, EFT matching computations
Amplitude action relation,
Pick your favorite formalism.

↓ **Straightforward**

2-Body Hamiltonian or Observables

Deal With 5PM Integration in Stages



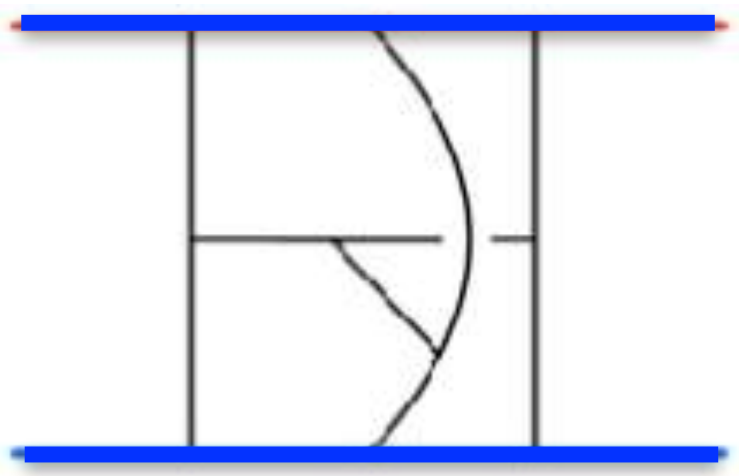
Stages:

1. **QED warmup. Potential mode contributions.** **Done.**
   ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng
2. **1SF Conservative + Disspative** **Done.**
   Driesse, Jakobsen, Mogull, Plefka, Sauer, Usovitsch
3. **$N = 8$ sugra (lower tensor rank)** **Both 1SF and 2SF potential done.**
   ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng
4. **GR 2SF "Conservative".** **Done**
   ZB, Herrmann, Roiban, Ruf, Smirnov, Smith, Zeng;
   Driesse, Jakobsen, Mogul, Nega, Plefka, Sauer, Usovitch
5. **Dissipative effects.** **To be done**

$$M_{5\mathrm{PM}} = M_{5\mathrm{PM}}^{0\mathrm{SF}} + \nu M_{5\mathrm{PM}}^{1\mathrm{SF}} + \nu^2 M_{\mathrm{PM}}^{2\mathrm{SF}} \qquad \nu = \frac{m_1 m_2}{(m_1+m_2)^2}$$

Learn from each stage to push forward the next one.

Bottleneck: Integral Evaluation



Primary bottleneck is integrals:

22 indices: 13 propagator and 9 irreducible scalar products (ISPs)

$$\int \frac{\mathrm{d}^{4D}k\,[u_2\cdot k_1]^{a_{-14}}[u_2\cdot k_4]^{a_{-15}}[u_1\cdot k_2]^{a_{-16}}[u_1\cdot k_3]^{a_{-17}}[k_1\cdot q]^{a_{-18}}[k_2\cdot q]^{a_{-19}}[k_1\cdot k_2]^{a_{-20}}[k_1\cdot k_4]^{a_{-21}}[k_2\cdot k_3]^{a_{-22}}}{[-2u_2\cdot k_2]^{a_1}[-2u_2\cdot k_{123}]^{a_2}[2u_1\cdot k_{234}]^{a_3}[2u_1\cdot k_{1234}]^{a_4}[k_1^2]^{a_5}[k_2^2]^{a_6}[k_3^2]^{a_7}[k_{13}^2]^{a_8}[k_4^2]^{a_9}[k_{34}^2]^{a_{10}}[k_{234}^2]^{a_{11}}[(k_{123}-q)^2]^{a_{12}}[(k_{1234}-q)^2]^{a_{13}}}$$

Very similar to wordliine integrals. KMOC version gives identical integrand as WLQFT

Capatti and Zeng

Encounter up to 8 numerator powers and 4 doubled propagators.

ZB, Herrmann, Roiban, Ruf, Smirnov,Smith, Zeng; Driesse, Jakobsen, Mogull, Plefka, Sauer, Usovitsch

The fundamental issue isn't the millions of integrals, it's the complexity.

No public code out of the box can deal with this.

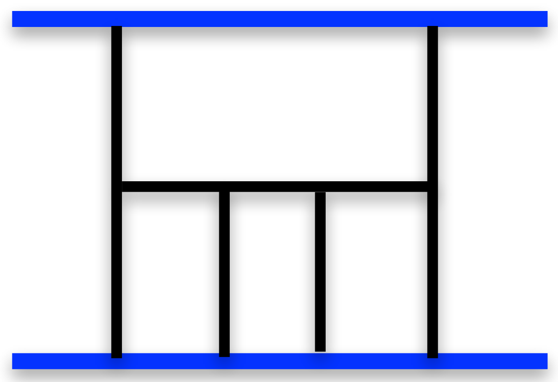
Basic Tool Is IBP

In QCD/Amplitudes advanced technology for loop integrals which we import.

1. **IBP greatly simplified in classical limit.** Chetyrkin, Tkachov; Laporta; Henn; Henn and Smirnov Beneke and Smirnov; Parra-Martinez, Ruf, Zeng

2. **Choose master integrals carefully to simplify IBPs.** A. Smirnov and V. Smirnov; Usovitsch

3. **Use finite prime fields and reconstruction for toughest IBPs.** Manteuffel, Schabinger; Peraro



Kotikov, ZB, Dixon and Kosower; Gehrmann, Remiddi

IBP:

$$0 = \int \prod_i^L \frac{d^D \ell_i}{(2\pi)^D} \frac{\partial}{\partial \ell_i^\mu} \frac{N^\mu(\ell_k, p_M)}{Z_1 \ldots Z_n}$$

Solve linear relations for master integrals, usually the simplest ones.

Our basic tools:

1. ***Highly modified* private version of FIRE.** Smirnov, Chuharev; Lee; Smirnov, Zeng
2. **Finite fields and reconstruction.** Von Manteuffel and a Schabinger; Peraro

KIRA has also be been upgraded. Maierhoefer, Usovitsch, Uwer; Klappert, Lange, Maierhöfer, Usovitsch

All integrals reduced to "Master integrals"

Finite Field Methods and Supercomputers

Question: **Given a large linear system (million unknowns) whose solution is rational functions over multiple variables, how to solve analytically?**

Answer: Finite field reconstruction.

A. von Manteuffel and R. Schabinger; T. Peraro

- **Problem reduced to numerical evaluation on a prime field.**
- **Rational coefficients of master integrals fully reconstructed.**

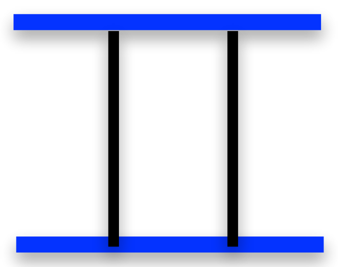
Trivially parallelizable.

Scale of the calculation requires supercomputers.

Supercomputers critical but better algorithms even more critical

DEs for Master Integrals.

IBP can also be used to generate differential equations:



$$\frac{\partial}{\partial y} I[1,1,1,1] \overset{\text{Lorenz-invariance}}{=} \frac{(u_2 - y u_1)}{1-y^2} \cdot \frac{\partial}{\partial u_1} I[1,1,1,1]$$

$$= \frac{y}{1-y^2} I[1,1,1,1] + \frac{1}{1-y^2} I[1,1,2,0]$$

$$\overset{\text{IBP}}{=} \frac{y}{1-y^2} I[1,1,1,1] + \frac{3-d}{1-y^2} I[1,1,0,0]$$

$$y = u_1 \cdot u_2$$

The DEs can be solved either analytically or by series expansion.

Structure of DEs

1st order coupled ODE

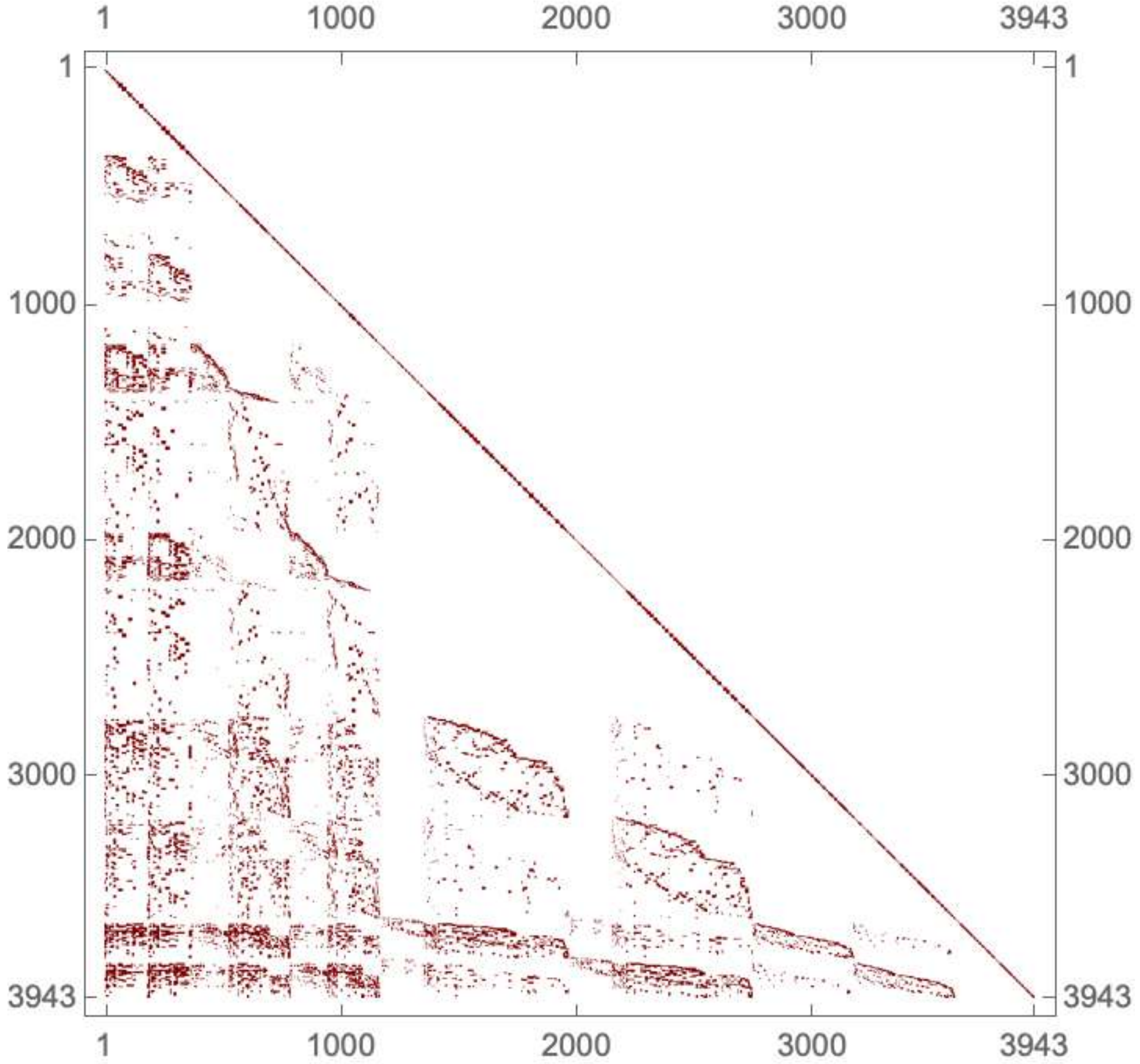
$m \times m$ matrix with rational entries

$$\frac{\partial}{\partial y}\begin{bmatrix} I_1 \\ \vdots \\ I_m \end{bmatrix} = M(y,\epsilon)\begin{bmatrix} I_1 \\ \vdots \\ I_m \end{bmatrix}$$

$D = 4 - 2\epsilon$

It takes considerable effort to clean up *M*.
Want “canonical form”

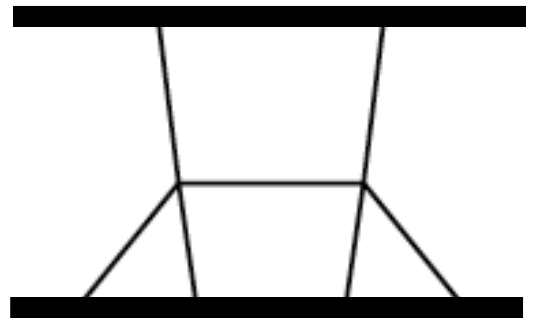
A 5 PM example of the DE matrix



Nontrivial Special Functions

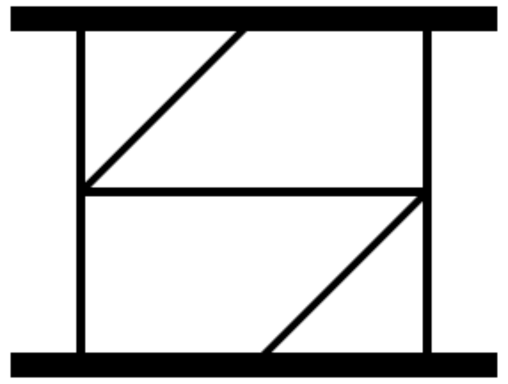
Elliptic, Heun and Calabi-Yau integrals appear.

Driesse, Jakobsen, Klemm, Mogull, Nega Plefka, Sauer, Usovitsch
ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Smirnov, Zeng.



At 1 SF, Elliptic integrals.

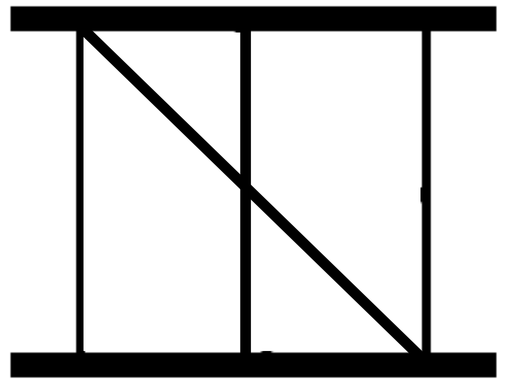
$$\Gamma_{i_1,\ldots,i_n}(x) = \int_1^x \mathrm{d}x_n\, k_{i_n}(x_n) \cdots \int_1^{x_2} \mathrm{d}x_1\, k_{i_1}(x_1)$$



At 2 SF, Calabi-Yau 3-fold integrals.

Complicated IBP and DEs

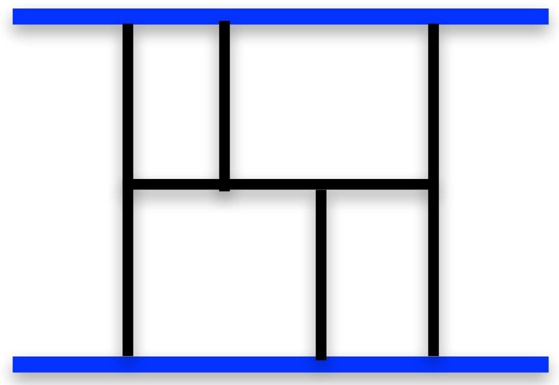
Frellesvig, Morales, Wilhelm; Klemm, Nega, Sauer, Plefka;
Driesse, Jokobsen, Klemm, Mogul, Nega, Pelfka, Sauer, Usovitsch;
Frellesvig, Morales, Pogel, Weinzierl, Wilhelm;
ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng;
ZB, Herrmann, Roiban, Ruf, Smirnov, Smith, Zeng;



At 2 SF, Heun type integral.

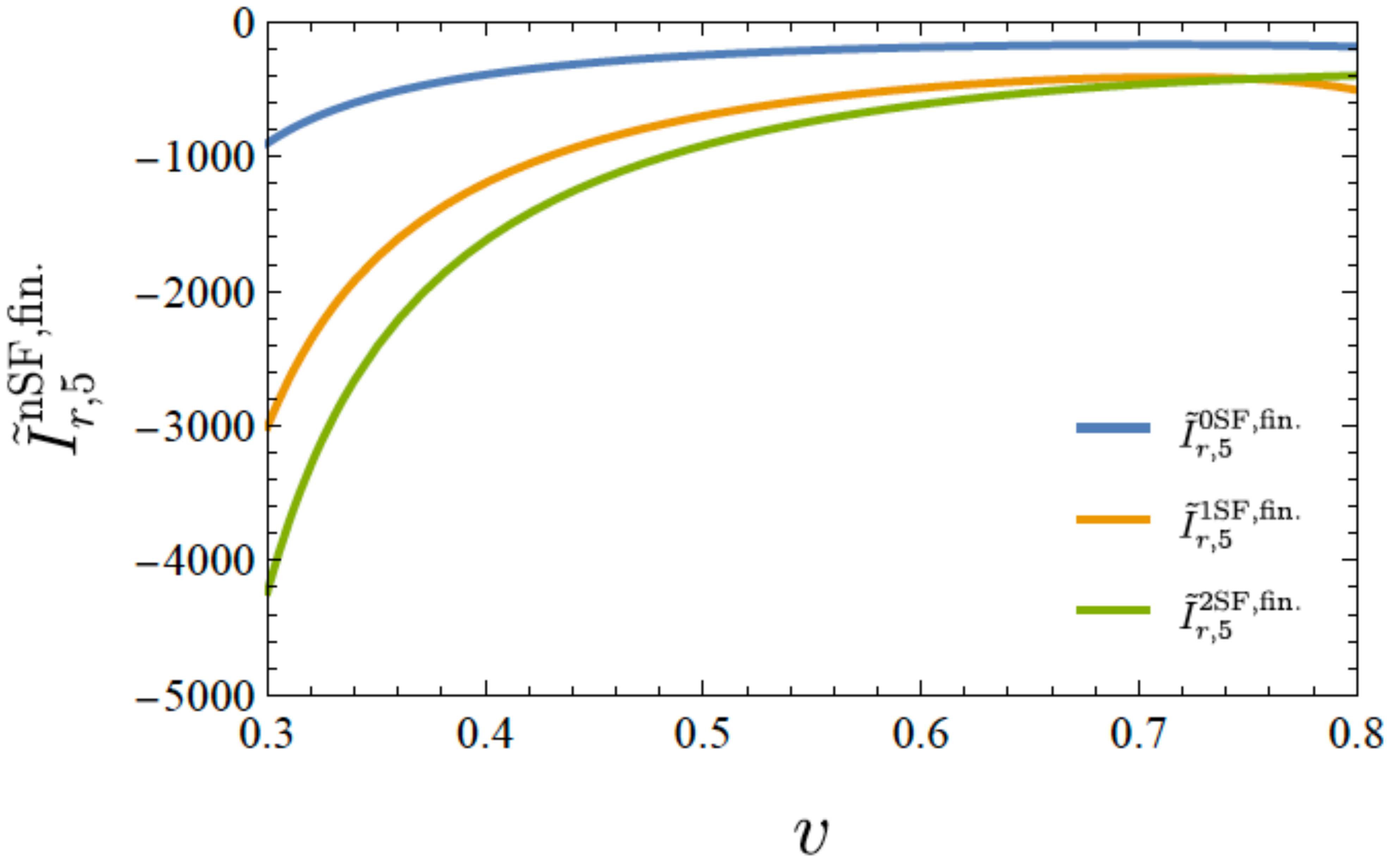
5PM GR Radial Action (Full 2SF Potential)

ZB, Herrmann, Roiban, Ruf, Smirnov, Smith, Zeng



$$\tilde{I}_{r,5} = m_1^2 m_2^2 (m_1^4 + m_2^4) \tilde{I}_{r,5}^{0\mathrm{SF}} + m_1^3 m_2^3 (m_1^2 + m_2^2) \tilde{I}_{r,5}^{1\mathrm{SF}} + m_1^4 m_2^4 \tilde{I}_{r,5}^{2\mathrm{SF}}$$

- **While Calabi-Yau drops out they complicate the solution.**
- **We know it drops out because the boundary condition from DE drops out.**
- **Easier to use series solution**



Scattering angle: $\chi = -\dfrac{\partial I_r}{\partial J}$

$$\tilde{I}_{r,5}^{2\mathrm{SF,fin.}} = -\frac{3}{320\bar{\sigma}^4} + \frac{31}{160\bar{\sigma}^3} - \frac{3543}{640\bar{\sigma}^2} - \frac{142651}{1920\bar{\sigma}} - \frac{22066709}{230400} + \dots$$
$$- \pi^2 \left(\frac{41}{64\bar{\sigma}} + \frac{9523}{1152} + \frac{44005\bar{\sigma}}{2304} + \dots \right)$$

22 terms explicitly given in paper

Plefka et al:
analytic expressions + conservative radiative part
A good way to separate conservative from dissipative is still lacking.

The way Elliptic and Calabi-Yau integrals drop out seems mysterious.

Can we get to 6PM?

More immediate question: Can we banish supercomputers at 5PM?

- Very fast progress though 4 PM (~1 day modest cluster, public codes).
  - No Integration bottleneck. Many order of magnitude simpler than 5PM.
  - FIRE6 or KIRA2 available for collider physics scattering amplitudes
- Progress slowed at 5PM because integrals beyond ability of public IBP codes.
  - — Upgraded IBP programs: FIRE7 and KIRA3, revamped alogorithms, better seeding.

FIRE7: Smirnov, Zeng; KIRA3: Lang, Usovitsch, Wu

Progress feeds back to collider physics!

Recent progress:

1) Better IBP seeding using mass dimension cutoff.
   — Targeted "bottleneck integrals" become "simple".
   — Much more progress on seeding possible.

2) Latch onto UV and IR properties. Example of a completely different approach.

Banishing supercomputers seems feasible.

Example of New Approach

ZB, Jackman, Mansfield, Ruf

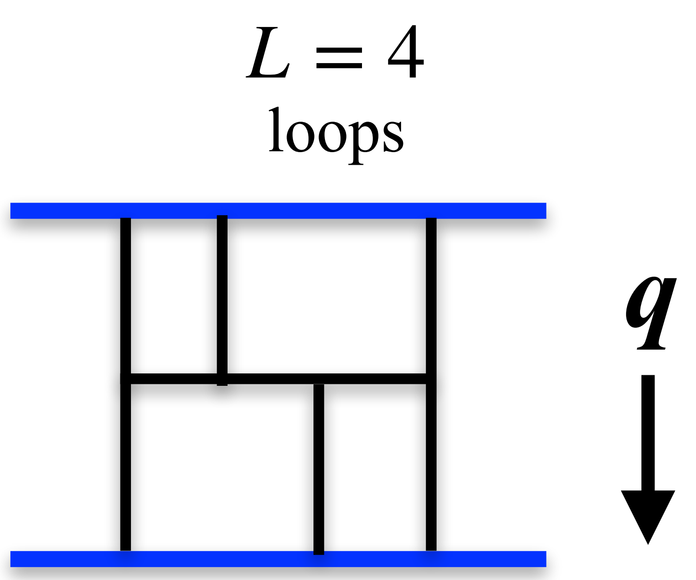
Why does elliptic cancel and CY cancel? Complete mystery via IBP.
There should be a better way to do the integrals!

In the conservative sector the scattering amplitudes at *L* loops behave as

$$\mathcal{M}_{\rm cl} \sim \begin{cases} (-q^2)^{\frac{L}{2}-1} \ln(-q^2)\,, & L \text{ even} \\ (-q^2)^{\frac{L}{2}-1}\,, & L \text{ odd}\,. \end{cases}$$

***q* is the momentum transfer**

$$V(r) \sim \frac{1}{r^{L+1}}$$



In dim. reg, the only source of ln(-q^2) are divergences

divergence:

$$\frac{1}{\epsilon}(-q^2)^{-L\epsilon} = \left[\frac{1}{\epsilon} - L\ln(-q^2) + \mathcal{O}(\epsilon)\right]$$

$$D = 4 - 2\epsilon$$

- **Diverge structure is *completely ignored* in the standard IBP approach.**
- **IBP can scramble IR and UV and introduces spurious divergences.**

Key observation: Singular regions generally much easier to calculate than nonsingular regions

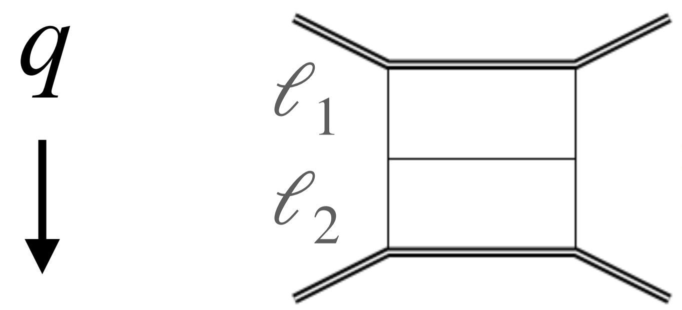
Exploiting UV Structure

Test of the idea:

1) Reproduced 2 loop (3PM) conservative results.   IBP methods not challenging here.

2) Can we explain missing Calabi-Yau and Elliptic integrals at 5PM?

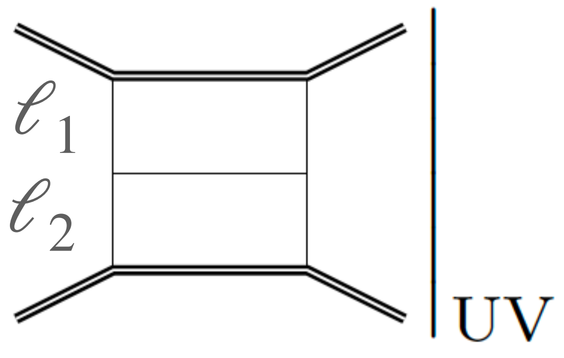
Two loop (3PM) example:



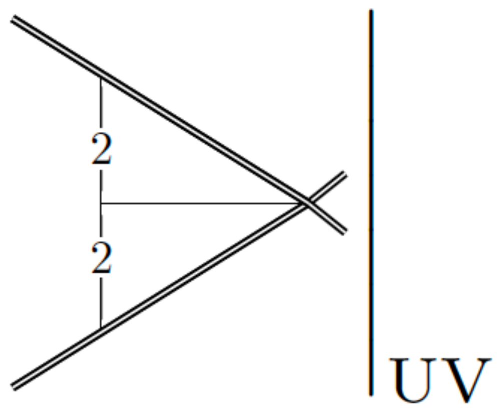
$$= \int \frac{\mathrm{d}^D\ell_1 \mathrm{d}^D\ell_2}{(\mathrm{i}\pi^{D/2})^2} \frac{\hat{\delta}(2u_1 \cdot \ell_1)\,\hat{\delta}(2u_2 \cdot \ell_2)\,\mathcal{N}}{\ell_1^2\ell_2^2(\ell_1-q)^2(\ell_2-q)^2(\ell_1+\ell_2-q)^2}$$

UV region

$\ell_1, \ell_2 >> q$



$=$



$$= \frac{\mathrm{i}\pi}{\epsilon}\frac{\operatorname{arccosh}(\sigma)(\mathrm{i}\pi - \operatorname{arccosh}(\sigma))}{2\sqrt{\sigma^2-1}}$$

UV divergence

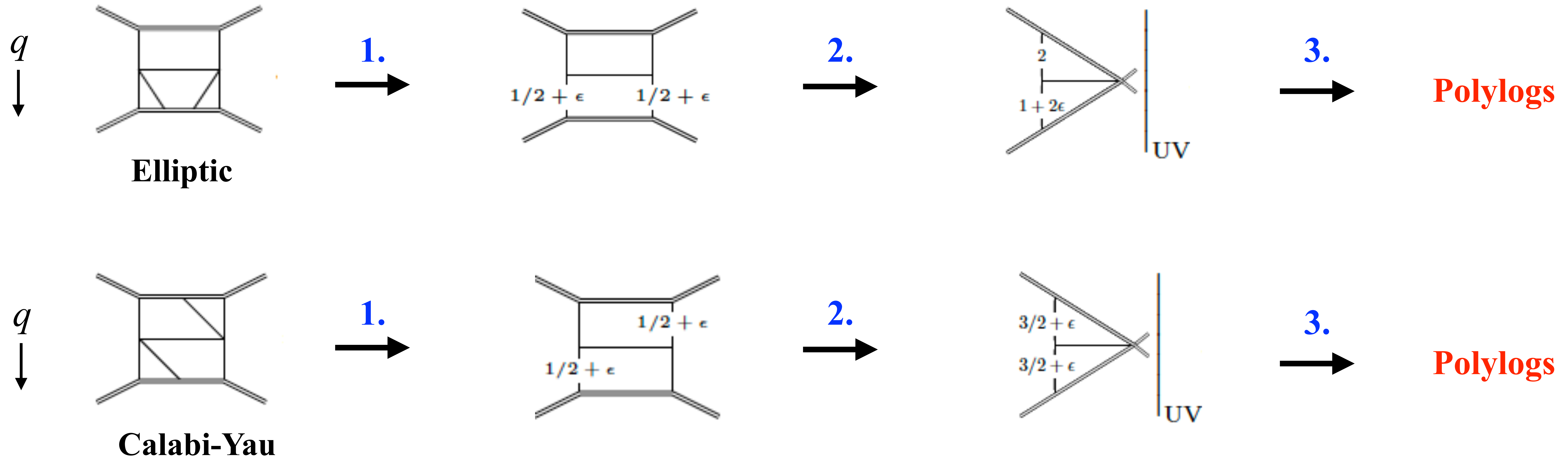
$(\ell_1 + q)^2 \to \ell_1^2$

- **In UV region integral simplifies into a cusp integral.**
- **Integrals simpler.**
- **Reproduces 3PM results**

Integrals simplify greatly in the UV

$D = 4 - 2\epsilon$

1. **Integrate out triangle subdiagrams.**
2. **Take UV limit:** $\ell_i \gg q$. $\quad (\ell_1 + q)^2 \to \ell_1^2$
3. **Integrate simpler UV integral (See Appendix of arXiv:)**



We see that elliptic and CY not present in conservative part. (Heun remains).

We are currently seeing how far this can be pushed.

Many Aspects Not Discussed Here

I only discussed spinless, no tidal, no absorption. Other aspects:

- **Pushing state of the art for high orders in *G***
  ZB, Cheung, Roiban, Parra-Martinez, Ruf, Shen, Solon, Zeng; ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Smith, Zeng; Damgaard, Plante, Vanhove; etc
- **Waveforms** Cristofoli, Gonzo, Kosower, O'Connell; Herrmann, Parra-Martinez, Ruf, Zeng; Di Vecchia, Heissenberg, Russo, Veneziano; Herderschee, Roiban, Teng; Georgoudis, Heissenberg , Vazquez-Holm; Brandhuber, Brown, Chen, De Angelis, Gowdy. Travaglini; De Angelis, Gonzo, Novichkov, Alaverdian, ZB, Kosmopoulos, Luna, RR, Scheopner, Teng; Bohnenblust, Ita, Kraus, Schlenk; Brunello, De Angelo, Kosower; etc
- **Finite-size effects** Cheung and Solon; Haddad and Helset; Kälin, Liu, Porto; Cheung, Shah, Solon; ZB, Parra-Martinez, Roiban, Sawyer, Shen; etc
- **Spin** Vaidya; Porto; Levi, Steinhoff; Geuvara, O'Connell, Vines; Chung, Huang, Kim, Lee; ZB, Luna, Roiban, Shen, Zeng; Mogull, Plefka, Steinhoff; Kosmopoulos, Luna; Febres Cordero, Kraus, Lin, Ruf, Zeng; Aoude Haddad, Helset; ZB, Kosmopoulos, Luna, Roiban, Scheopner, Teng, Vines; Cangemi, Chiodaroli, Johansson, Ochirov, Pichini, Skvortsov; Gatica; Luna, Moynihan, O'Connell, Ross; Akpinar, Febres Cordero, Kraus, Ruf, Zeng; Bautista, Guevara, Kavanagh, Vines; etc.
- **Absorption** Jones and Ruf, Chen, Hsieh, Y-t Huang, J-W Kim; Bautista, Guevara, Kavanagh, Vines; etc
- **Renormalization Effects** Goldberger and Rothstein; Goldberger, Ross; Porto, Galley, Leibovich; Levi; Barack, ZB, Herrmann, Long, Parra-Martinez, Roiban, Ruf, Shen, Solon, Teng, Ivanov, Li, Parra-Martinez, Zhou; Caron-Huot, Correia, Isabella, Solon, etc

The standard quantities of interest for the inspiral phase of binary black holes can be computed in amplitudes approach. A lot of work remains!

Summary

- Pushing state of the art: real progress on 5PM . "Conservative" part under control.

- Progress is slower now that we are beyond reach standard IBP tools.

- New ideas and methods for tackling integrals

  — Better IBP seeding. Recent vast improvements. More possible.

  — UV, IR analysis is one new idea, at least for certain classes of integrals, including 5PM "conservative".

  — Elliptic and CY integral cancellations explained.

- 5PM radiative, dissapative contributions under study.

- Longer term goal is 6PM. First need to banish supercomputers from 5PM.

5PM is difficult but coming under control.

EXTRA

Systematically Correcting Newton's Potential

For orbital mechanics:

Expand in G and v^2

$$v^2 \sim \frac{GM}{R} \ll 1$$

virial theorem

In center of mass frame:

$$m = m_A + m_B; \quad \nu = \mu/M,$$

$$\mu = m_A m_B/m; \quad P_R = P \cdot \hat{R}$$

$$\frac{H}{\mu} = \frac{P^2}{2} - \frac{Gm}{R}$$

Newton

$$+ \frac{1}{c^2}\left\{ -\frac{P^4}{8} + \frac{3\,\nu\,P^4}{8} + \frac{Gm}{R}\left(-\frac{{P_R}^2\,\nu}{2} - \frac{3\,P^2}{2} - \frac{\nu\,P^2}{2}\right) + \frac{G^2m^2}{2R^2}\right\}$$

$$+ \ldots$$

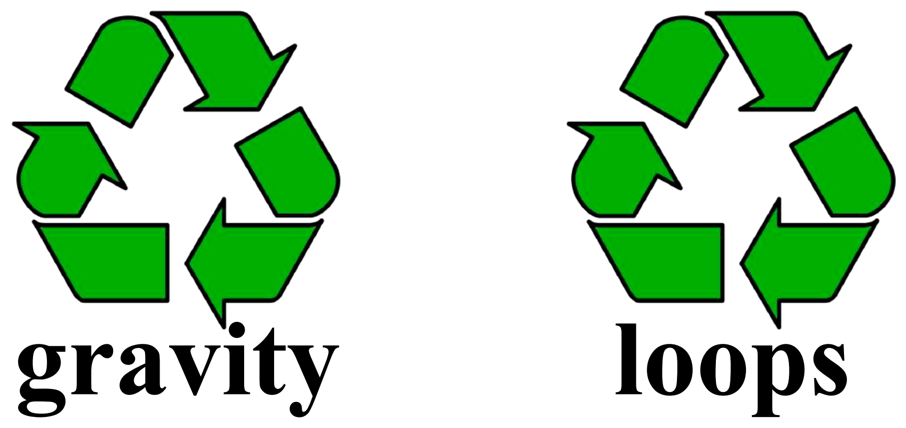
1PN: Einstein, Infeld, Hoffmann; Droste, Lorentz

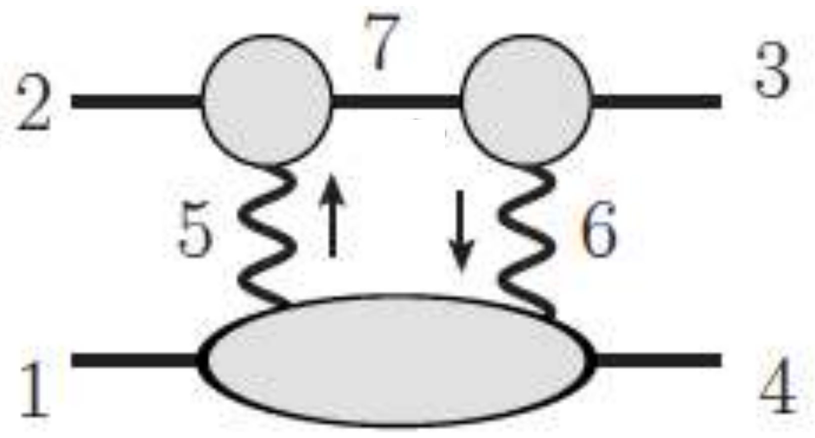
Hamiltonian known to 4PN order.

2PN: Ohta, Okamura, Kimura and Hiida.
3PN: Damour, Jaranowski and Schaefer; L. Blanchet and G. Faye.
4PN: Damour, Jaranowski and Schaefer; Foffa (2017), Porto, Rothstein, Sturani (2019).

Generalized Unitarity Cuts

- Higher-order gravity (loops) from lower order (tree) amplitudes.
- Gravity tree amplitudes from gauge-theory ones.





2nd post-Minkowkian order

$$C_{\rm GR} = \sum_{h_5,h_6=\pm} M_3^{\rm tree}(3^s, 6^{h_6}, -7^s)\, M_3^{\rm tree}(7^s, -5^{h_5}, 2^s)\, M_4^{\rm tree}(1^s, 5^{-h_5}, -6^{-h_6}, 4^s)$$

gravity tree amplitudes

$$= \sum_{h_5,h_6=\pm} it[A_3^{\rm tree}(3^s, 6^{h_6}, -7^s)\, A_3^{\rm tree}(7^s, -5^{h_5}, 2^s)\, A_4^{\rm tree}(1^s, 5^{-h_5}, -6^{-h_6}, 4^s)]$$
$$\times [A_3^{\rm tree}(3^s, 6^{h_6}, -7^s)\, A_3^{\rm tree}(7^s, -5^{h_5}, 2^s)\, A_4^{\rm tree}(4^s, 5^{-h_5}, -6^{-h_6}, 1^s)]$$

gauge theory amplitudes

- Note: Calculations start from gauge theory not gravity.
- This is then appropriately integrated and processed to give observable

EXTRA

Issues to consider.

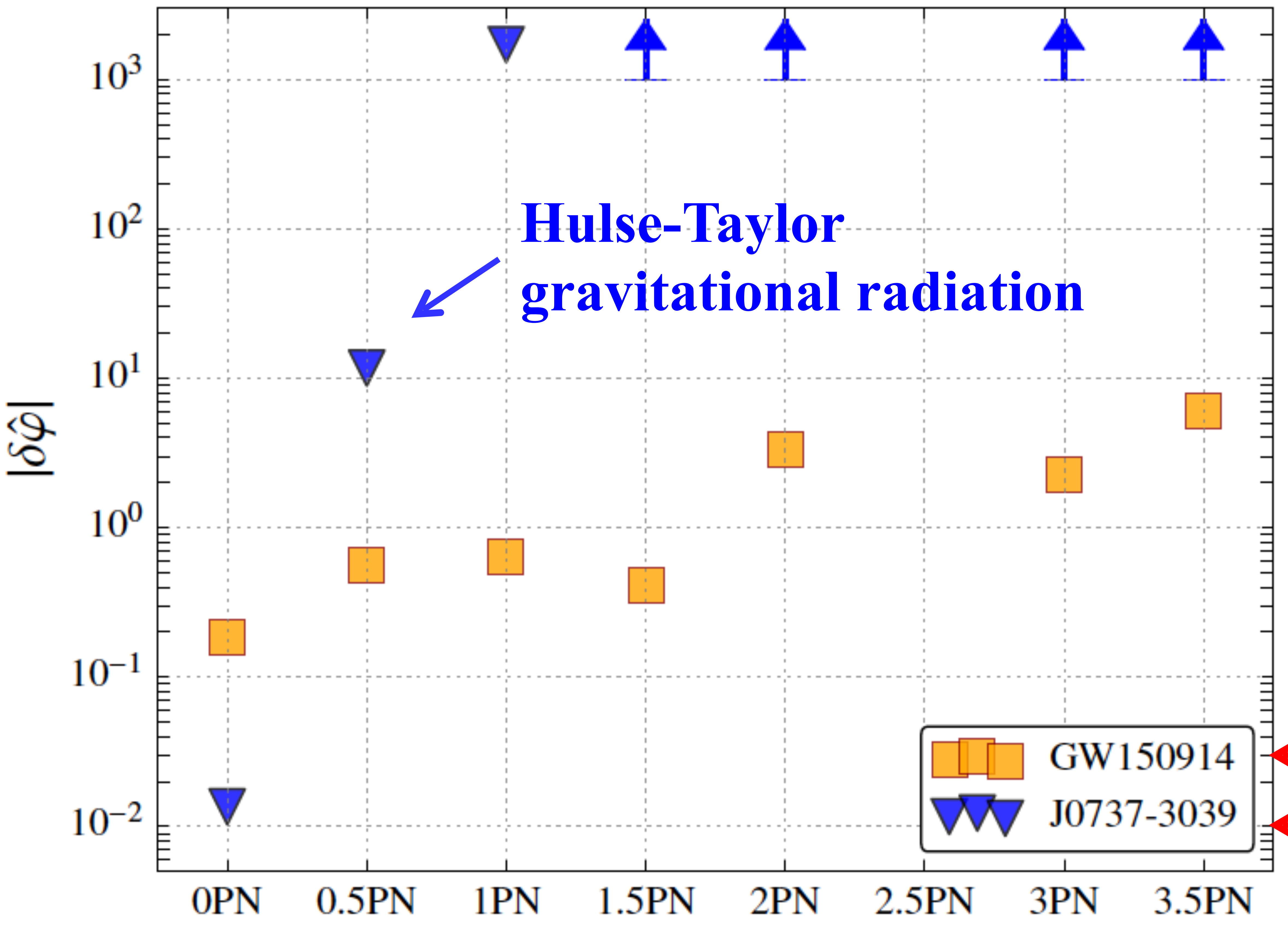
Should we keep the slide on the detectibility

Is there too much double copy.

Reduce emphasis on the last part.

Importance of higher orders for LIGO/Virgo

LIGO/Virgo Collaboration arXiv:1602.03841



Hulse-Taylor gravitational radiation

Newton (1686)

Einstein (1938)

Ohta (1974)
Damour (1981)

Blanchet (2000)
Damour (2000)

Blanchet (2015)
Damour (2014)

LIGO

Binary pulsar

Binary pulsar confirms quadrupole radiation formula and not much else.

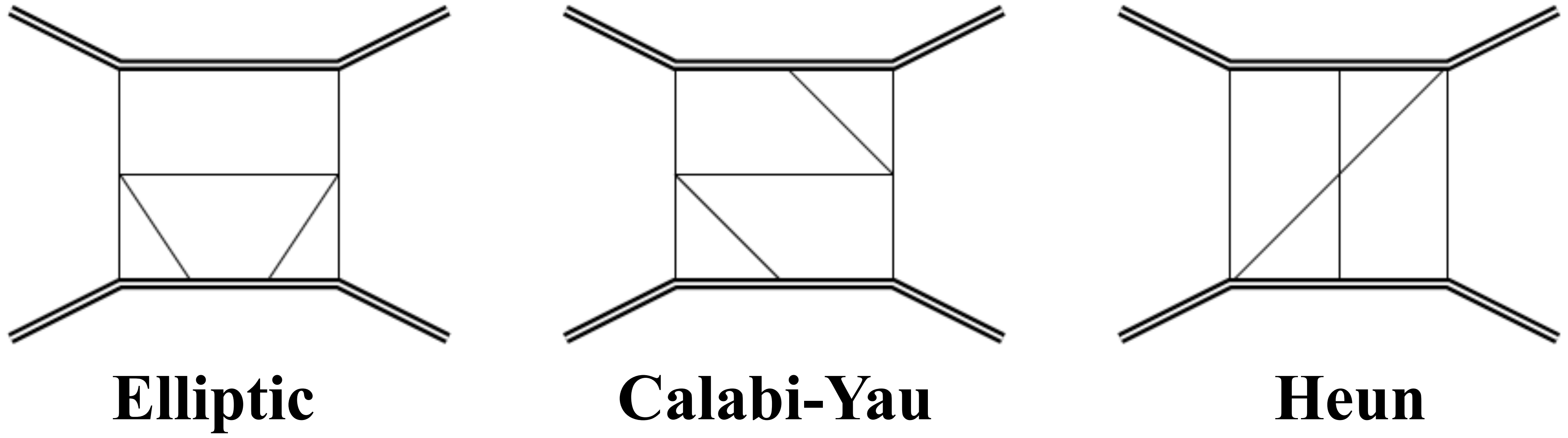
LIGO/Virgo tests PN terms from GR

LIGO/Virgo sensitive to high PN orders.

Source of Calabi-Yau, Elliptic and Heun

Integration facts we exploit:

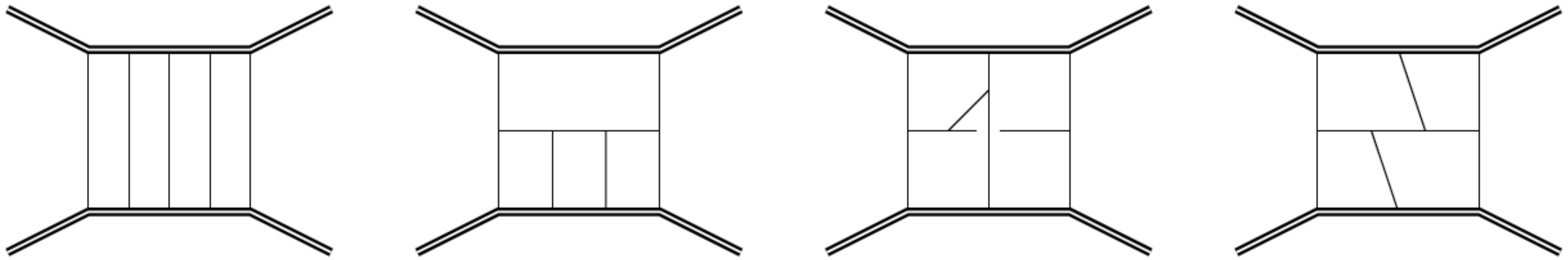
- The special functions beyond GPLs are controlled by only 3 sources:



These are the diagrams to focus on

- Above diagrams have only UV divergences, no IR

- Maximal cut analysis shows above diagrams must be present for functions beyond GPLS



Task: Determine what happens to the Elliptic, CY and Heun in the UV limit.

What are we after?

- Physical observables directly.
- When possible, replace scattering in general gelativity with a two-body Hamiltonian that is easy to use in bound-state problem.
- $$V(\mathbf{r},\mathbf{p}) = -\frac{G m_1 m_2}{r} + \cdots$$

want correction terms

Similar to Newton's potential, except:

- **Compatible with special relativity (all orders in velocity)**
- **Eventually valid through $O(G^6)$.**

To Meet Challenge: GWSky Project

Alessandra Buonanno
(AEI, Potsdam)

Gravitational-wave
modeling, interpretation

Zvi Bern (UCLA)

Collider physics,
black-hole scattering

Non-linear strong
field dynamics, radiation

Maarten van de Meent
Bohr Institute, Copenhagen

Black-hole environments,
beyond General Relativity

Enrico Barausse
SISSA, Trieste.

Funded by
the European Union

European Research Council
Established by the European Commission

PN versus PM expansion for conservative two-body dynamics

From Buonanno
Amplitudes 2018

$$\mathcal{L} = -Mc^2 + \underbrace{\frac{\mu v^2}{2} + \frac{GM\mu}{r}}_{\downarrow} + \frac{1}{c^2}\Big[\dots\Big] + \frac{1}{c^4}\Big[\dots\Big] + \dots$$

$$E(v) = -\frac{\mu}{2}v^2 + \cdots$$

non-spinning compact objects

		0PN	1PN	2PN	3PN	4PN	5PN	...
0PM:	1	v^2	v^4	v^6	v^8	v^{10}	v^{12}	...
1PM:		$1/r$	v^2/r	v^4/r	v^6/r	v^8/r	v^{10}/r	...
2PM:			$1/r^2$	v^2/r^2	v^4/r^2	v^6/r^2	v^8/r^2	...
3PM:				$1/r^3$	v^2/r^3	v^4/r^3	v^6/r^3	...
4PM:					$1/r^4$	v^2/r^4	v^4/r^4	...
...						...	...	

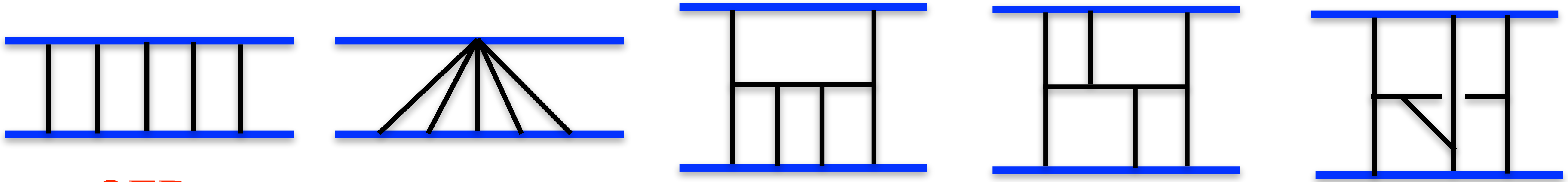
(credit: Justin Vines)

$$1 \to Mc^2, \qquad v^2 \to \frac{v^2}{c^2}, \qquad \frac{1}{r} \to \frac{GM}{rc^2}.$$

current known PN results — current known PM results — overlap between PN & PM results — unknown

- **PM results** (Westfahl 79, Westfahl & Goller 80, Portilla 79-80, Bel et al. 81, Ledvinka et al. 10, Damour 16-17, Guevara 17, Vines 17, Bini & Damour 17-18, Vines in prep)

Deal With Integration in Stages



QED 0SF 1SF 2SF 2SF

Stages:

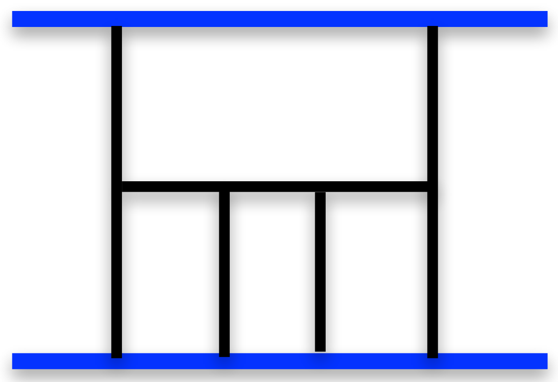
1. **QED warmup. Potential mode contributions.** **Done.**
   ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng
2. **1 SF Conservative.** **Done.**
   Driesse, Jakobsen, Mogull, Plefka, Sauer, Usovitsch
3. $\mathcal{N} = 8$ **sugra(lower tensor rank)** **Both 1SF and 2SF potential done.**
   ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng
4. **2 SF Conservative.** **Hard, but definitely in reach.**
5. **Radiative effects.** **Similar**

$$M_{5\mathrm{PM}} = M_{5\mathrm{PM}}^{0\mathrm{SF}} + \nu M_{5\mathrm{PM}}^{1\mathrm{SF}} + \nu^2 M_{\mathrm{PM}}^{2\mathrm{SF}} \qquad \nu = \frac{m_1 m_2}{(m_1+m_2)^2}$$

Learn from each stage to push forward the next one.

High Loop Integration

In QCD/Amplitudes advanced technology for loop integrals which we import.



1. **IBP greatly simplified in classical limit.** Chetyrkin, Tkachov; Laporta; Henn; Henn and Smirnov Beneke and Smirnov; Parra-Martinez, Ruf, Zeng
2. **Choose master integrals to simplify the DEs and IBP.** A. Smirnov and V. Smirnov; Usovitsch
3. **Use finite prime fields and reconstruction for toughest IBPs.** Manteuffel, Schabinger; Peraro
4. **Set up a DEs for master integrals.** Kotikov, ZB, Dixon and Kosower; Gehrmann, Remiddi

IBP: $$0 = \int \prod_i^L \frac{d^D \ell_i}{(2\pi)^D} \frac{\partial}{\partial \ell_i^\mu} \frac{N^\mu(\ell_k, p_M)}{Z_1 \dots Z_n}$$ **Solve linear relations for master integrals**

DEs: $$\partial_x \vec{I} = A(x,\epsilon)\vec{I}\,, \qquad \partial_x \vec{J} = \epsilon\, B(x)\vec{J}\,, \qquad \vec{J} = T(\epsilon, x)\vec{I}$$ **Solve DEs either as series or basis of functions.**

Many tools available: We use upgraded FIRE, a private finite field IBP program, LiteRed, FiniteFlow.

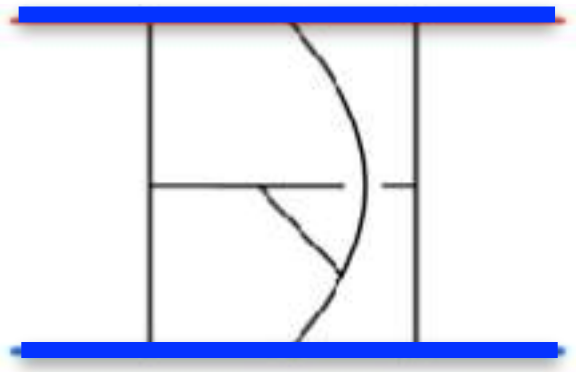
Smirnov, Chuharev; Lee; Peraro

Another important tool is an upgraded version of KIRA

Maierhoefer, Usovitsch, Uwer;
Klappert, Lange, Maierhöfer, Usovitsch

Bottleneck: Integral Reduction

Primary bottleneck is integration by parts:



22 indices: 13 propagator and 9 irreducible scalar products (ISPs)

$$\int \frac{\mathrm{d}^{4D}k\, [u_2 \cdot k_1]^{a_{-14}} [u_2 \cdot k_4]^{a_{-15}} [u_1 \cdot k_2]^{a_{-16}} [u_1 \cdot k_3]^{a_{-17}} [k_1 \cdot q]^{a_{-18}} [k_2 \cdot q]^{a_{-19}} [k_1 \cdot k_2]^{a_{-20}} [k_1 \cdot k_4]^{a_{-21}} [k_2 \cdot k_3]^{a_{-22}}}{[-2u_2 \cdot k_2]^{a_1} [-2u_2 \cdot k_{123}]^{a_2} [2u_1 \cdot k_{234}]^{a_3} [2u_1 \cdot k_{1234}]^{a_4} [k_1^2]^{a_5} [k_2^2]^{a_6} [k_3^2]^{a_7} [k_{13}^2]^{a_8} [k_4^2]^{a_9} [k_{34}^2]^{a_{10}} [k_{234}^2]^{a_{11}} [(k_{123}-q)^2]^{a_{12}} [(k_{1234}-q)^2]^{a_{13}}}$$

Very similar to wordliine integrals. KMOC version gives identical integrand as WLQFT

- **Can encounter up to 8 numerator powers and 4 doubled propagators.**
- **FIRE and KIRA need to be carefully tuned.**
- **Private code specialized for finite field numerical techniques.**

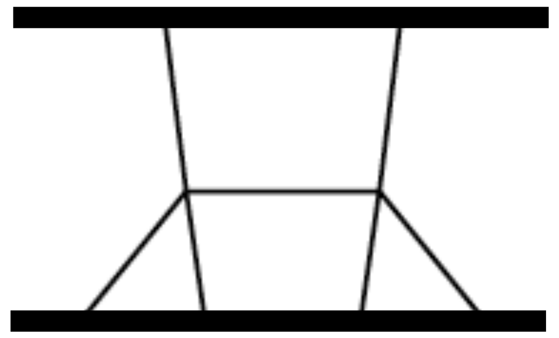
$$0 = \int \prod_i^L \frac{d^D \ell_i}{(2\pi)^D} \frac{\partial}{\partial \ell_i^\mu} \frac{N^\mu(\ell_k, p_i)}{D_1^{a_1} \cdots D_n^{a_n}}$$

ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng; Driesse, Jakobsen, Mogull, Plefka, Sauer, Usovitsch

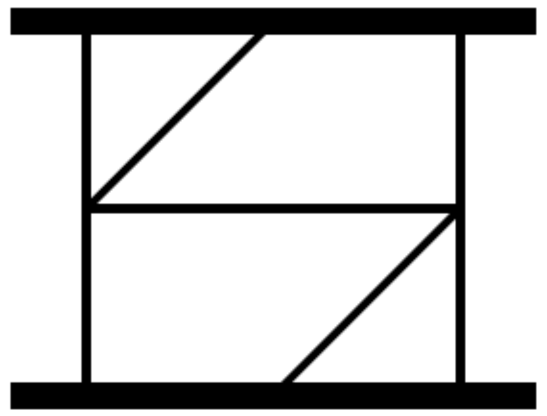
Iterated integrals

Elliptic and Calabi-Yau integrals appear in master integrals.

Frellesvig, Morales, Wilhelm; Klemm, Nega, Sauer, Plefka
ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng.



At 1 SF, Elliptic integrals.



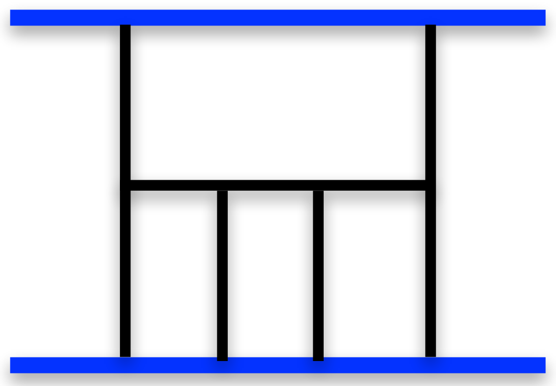
At 2 SF, Calabi-Yau 3-fold integrals.

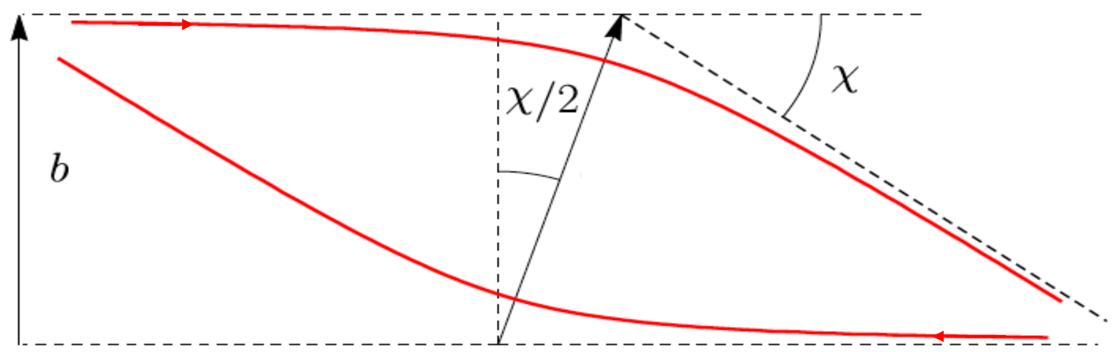
- **At 1SF no elliptic integrals in final result!**
- **Will this simplicity continue to 2 SF sector?**

Opportunity for those interested in the mathematics of Feynman integrals

5PM Scattering Angle $N = 8$ Supergravity (1SF Potential)

ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng





$$\tilde{I}_{r,5}^{\text{1SF,fin.}} = r_1 + r_2\,\mathrm{F}_0 + r_3\,\mathrm{F}_0^2 + r_4\,\mathrm{F}_1 + r_5\,\mathrm{F}_2\,.$$

$$\mathrm{F}_0 = \frac{2x}{1-x^2}\ln(x)\,,$$

$$\mathrm{F}_1 = \frac{2x}{1-x^2}\left[-\mathrm{Li}_2(1-x) - \mathrm{Li}_2(-x) - \ln(x)\ln(x+1) - \frac{1}{2}\zeta_2\right],$$

$$\mathrm{F}_2 = \frac{2x}{1-x^2}\left[-\mathrm{Li}_2(1-x) + \mathrm{Li}_2(-x) - \frac{1}{2}\ln^2(x) + \ln(x)\ln(x+1) + \frac{1}{2}\zeta_2\right]$$

Cyclotomic polylogs are natural functions to use, but here only up to dilogs. Ablinger, Bluemlein, Schneider.

Remarkably simple, elliptic integrals cancel.

$$r_1 = \frac{16c_\phi^3}{\sigma^2-1}\left[-\frac{\sigma(5\sigma^2-4)c_\phi^3}{5(\sigma^2-1)^3} - \frac{2c_\phi^2}{\sigma^2-1} + 8\right],$$

$$r_2 = 32c_\phi^2\left[-\frac{\sigma^2 c_\phi^4}{(\sigma^2-1)^3} - \frac{4\sigma c_\phi^3}{(\sigma^2-1)^2} - \frac{9(2\sigma^2-1)c_\phi^2}{2(\sigma^2-1)^2} - \frac{8\sigma c_\phi}{\sigma^2-1} + 1\right],$$

$$r_3 = 16c_\phi\left[-\frac{\sigma^3 c_\phi^5}{(\sigma^2-1)^3} - \frac{6\sigma^2 c_\phi^4}{(\sigma^2-1)^2} + \frac{6\sigma(2-3\sigma^2)c_\phi^3}{(\sigma^2-1)^2} + \frac{(8-32\sigma^2)c_\phi^2}{\sigma^2-1} - \frac{92\sigma c_\phi}{3} - \frac{40}{3}(\sigma^2-1)\right],$$

$$r_4 = 32c_\phi\left[-\frac{\sigma c_\phi^3}{(\sigma^2-1)^2} - \frac{2c_\phi^2}{\sigma^2-1} - \frac{4\sigma c_\phi}{3(\sigma^2-1)} - \frac{8}{3}\right],$$

$$r_5 = 64c_\phi\left[\frac{\sigma^2 c_\phi^3}{(\sigma^2-1)^2} + \frac{4\sigma c_\phi^2}{\sigma^2-1} + \frac{2(7\sigma^2-6)c_\phi}{3(\sigma^2-1)} + \frac{4\sigma}{3}\right].$$

See also very nice GR 1SF conservative results.

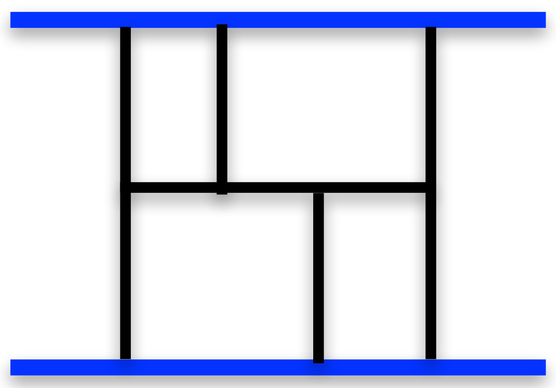
Driesse, Jakobsen, Mogull, Plefka, Sauer, Usovitsch

Scattering angle: $\chi = -\dfrac{\partial I_r}{\partial J}$

Very encouraging that results are so simple

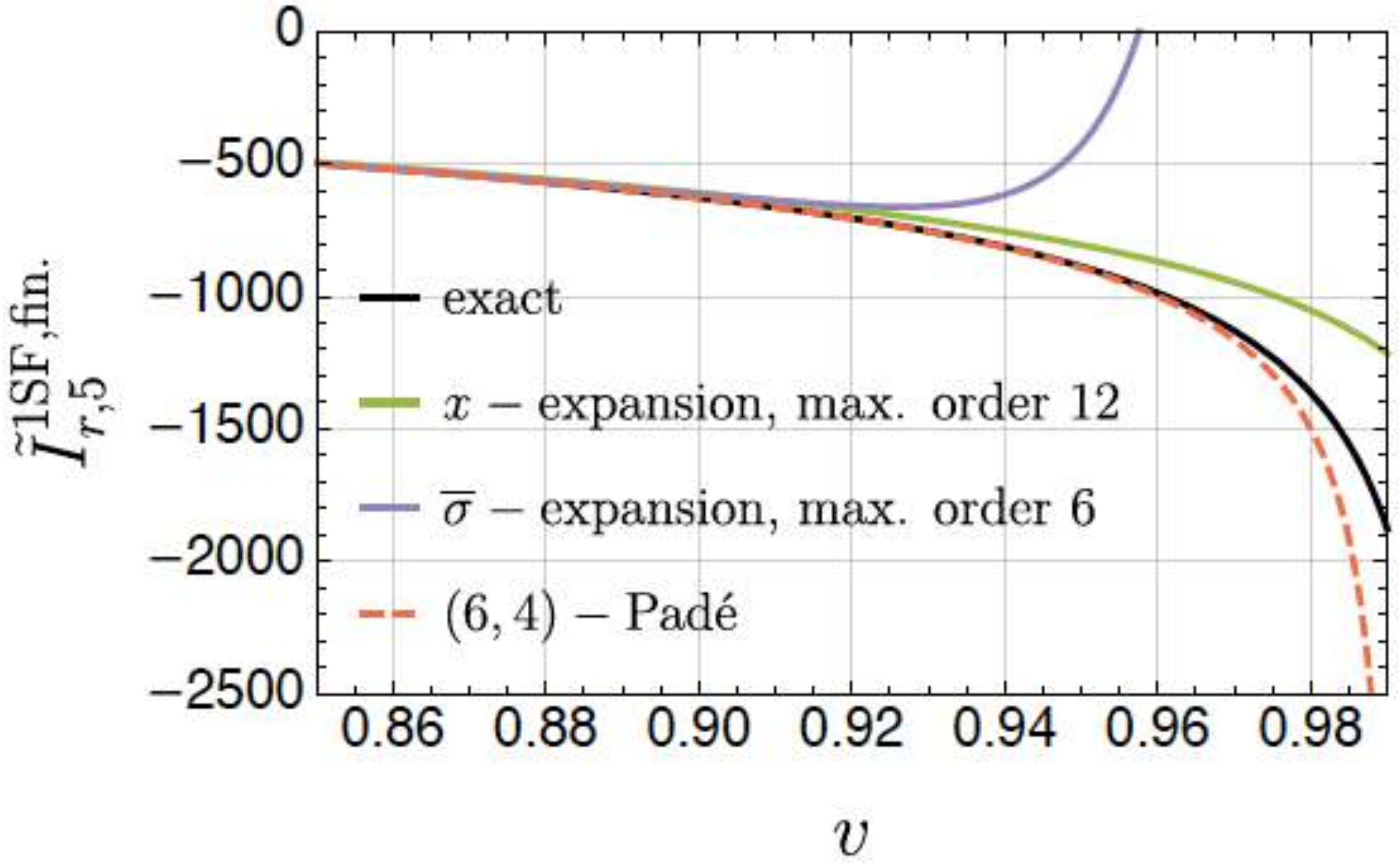
5PM Scattering Angle $N = 8$ Supergravity (2SF Potential)

ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng

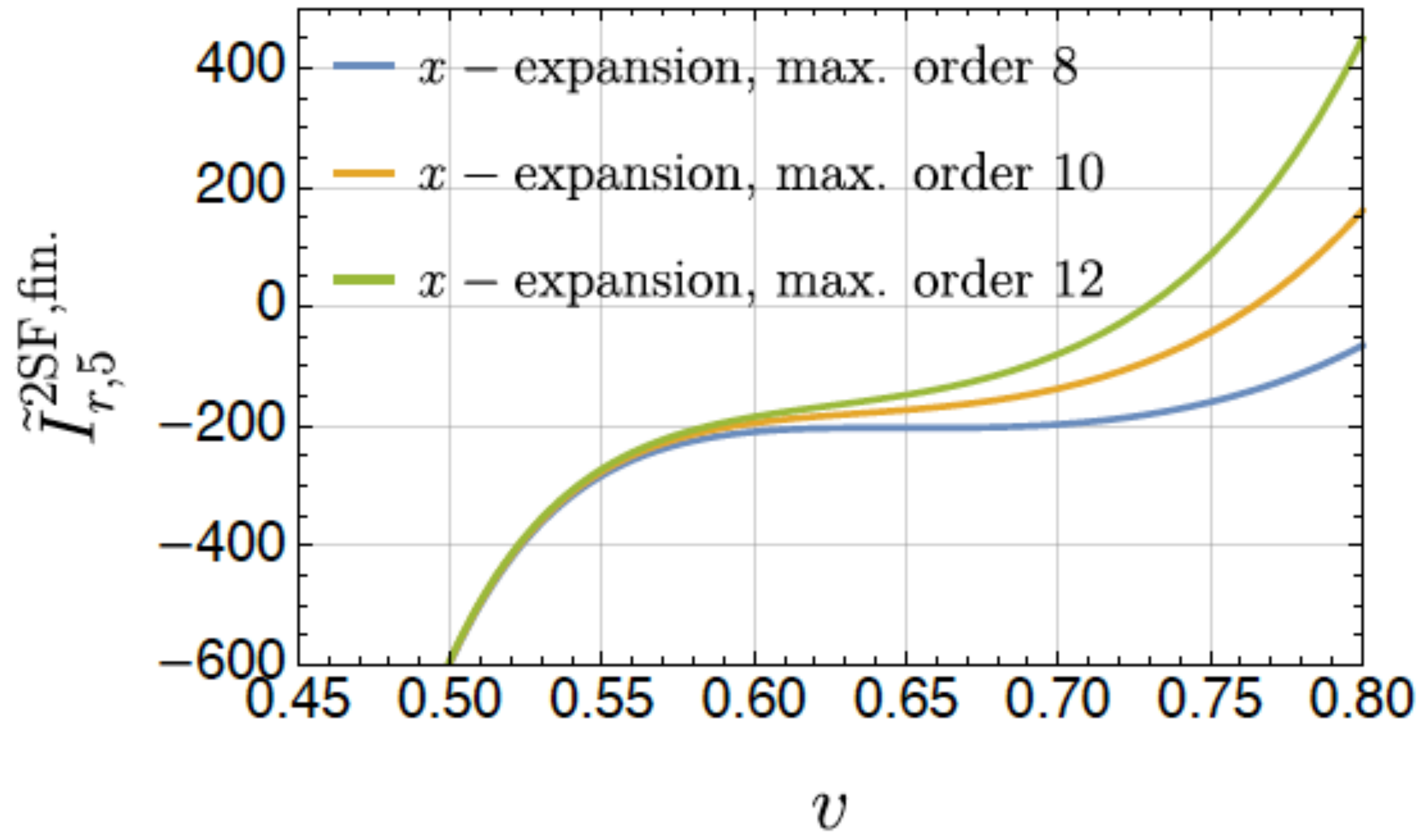


For 2SF Calabi-Yau integrals complicate analytic solution, use series solution

$$\tilde{I}_{r,5}^{\mathrm{2SF,fin.}} = 16c_\phi \Bigg[c_\phi^5 \left(-\frac{3}{160\bar{\sigma}^4} + \frac{1}{80\bar{\sigma}^3} + \cdots \right) + c_\phi^4 \left(-\frac{4}{3\bar{\sigma}} + \frac{38}{45} + \cdots \right) + c_\phi^3 \left(-\frac{121}{18\bar{\sigma}} - \frac{5933}{1800} + \cdots \right)$$
$$+ c_\phi^2 \left(-\frac{12}{\bar{\sigma}} - \frac{370}{9} + \cdots \right) + c_\phi^1 \left(-\frac{302}{3} + \frac{1016\bar{\sigma}}{45} + \cdots \right) + c_\phi^0 \left(-\frac{296\bar{\sigma}}{3} + \frac{76\bar{\sigma}^2}{5} + \cdots \right) \Bigg]$$



Worked out 10 orders



Expect series solutions should be fine for practical purposes

Outlook

Amplitude methods have already contributed to many aspects of theory of gravitational waves.

- **Pushing state of the art for high orders in *G*** ZB, Cheung, Roiban, Parra-Martinez, Ruf, Shen, Solon, Zeng; ZB, Herrmann, Roiban, Ruf, Smirnov, Smirnov, Zeng; Damgaard, Plante, Vanhove; etc
- **Waveforms** Cristofoli, Gonzo, Kosower, O'Connell; Herrmann, Parra-Martinez, Ruf, Zeng; Di Vecchia, Heissenberg, Russo, Veneziano; Herderschee, Roiban, Teng; Georgoudis, Heissenberg , Vazquez-Holm; Brandhuber, Brown, Chen, De Angelis, Gowdy. Travaglini; De Angelis, Gonzo, Novichkov, Alaverdian, ZB, Kosmopoulos, Luna, RR, Scheopner, Teng; Bohnenblust, Ita, Kraus, Schlenk; etc
- **Finite-size effects** Cheung and Solon; Haddad and Helset; Kälin, Liu, Porto; Cheung, Shah, Solon; ZB, Parra-Martinez, Roiban, Sawyer, Shen; etc
- **Spin** Vaidya; Geuvara, O'Connell, Vines; Chung, Huang, Kim, Lee; ZB, Luna, Roiban, Shen, Zeng; Kosmopoulos, Luna; Febres Cordero, Kraus, Lin, Ruf, Zeng; Aoude Haddad, Helset; ZB, Kosmopoulos, Luna, Roiban, Scheopner, Teng, Vines; Cangemi, Chiodaroli, Johansson, Ochirov, Pichini, Skvortsov; Akpinar, Febres Cordero, Kraus, Ruf, Zeng; Bautista, Guevara, Kavanagh, Vines; etc.
- **Absorption** Jones and Ruf, Chen, Hsieh, Y-t Huang, J-W Kim; Bautista, Guevara, Kavanagh, Vines; etc

The standard quantities of interest for the inspiral phase can be computed in amplitudes framework. Formalism is far from exhausted.

Summary

- **Scattering amplitudes give us new ways to think about certain problems of current interest in general relativity.**
- **Feynman's approach: massless spin-2 gravitons all you need to derive GR.**
- **Double-copy idea gives a *unified framework* for gravity and gauge theory.**
- **Combining with EFT methods gives a powerful tool for gravitational-wave physics.**
- **Pushing state of the art: heading to O(G^5).**
- **GWSky Project Goal: Combine with other methods to meet precision challenge.**
- **Various directions: higher orders in *G*, resummations in *G*, spin, finite-size effects, radiation and dissipation.**

Amplitudes viewpoint that gravity and gauge theory belong together is powerful.

Expect much more progress in the coming years.